



ALMA MATER STUDIORUM  
UNIVERSITÀ DI BOLOGNA

## LLM Benchmark project Report

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**Repository:** [https://github.com/albertov-4/LLM\\_Benchmark.git](https://github.com/albertov-4/LLM_Benchmark.git)

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# 1 Introduction

The main goal of this project is to provide a general benchmark to evaluate different LLMs performances under different types of planning domains to better understand in advance what are the planning capabilities of a given LLM. In the following sections the main features of this projects are going to be shown.

Also a quick PDDL starting guide has been created to better gain the necessary knowledge about PDDL this specific language used to define planning problems.

## 2 General Structure

The root folder contains various sub folders, specifically:

- analysis
- archive
- Benchmark Framework
- results

### 2.1 analysis

This folder is used to gather all the datas useful for the production fo the report and the slides, contains the summary of the raw scores for the PDDL domains and the jupiter notebooks and the report.

### 2.2 archive

This folder contains all the material given us by the professor, specifically papers, a list of PDDL planning domains, two github repos given us as a reference and last a Quick PDDL guide written by us to help understand better the language.

### 2.3 Benchmark Framework structure

The folder Benchmark Framework is the core of all the work we've done. The main goal of this framework was to provide a easy to use and general testing platform to be employed when evaluating the performances of a LLM is needed, The following pages are going to be used to show the folders and the structure of the framework:

```
LLM Benchmark/  
├── docs/  
├── evaluators/  
├── Leonardo script/  
├── models/  
├── outputs/  
├── prompts/  
├── protocols/  
├── reporting/  
├── runner/  
├── scripts/  
├── slurm logs/  
├── tasks/  
├── tests/  
├── utils/  
├── advanced planning evaluation.py  
├── clear outputs.py  
├── run benchmark.py  
└── setup.md
```

**docs** Contains a single file for the model preparation which is useful because the benchmark can run on a local PC or on an HPC system such as Leonardo. The important operational rule is that GPU benchmark jobs should not download large model weights. Prepare Hugging Face models first, then run benchmarks from local files.

**evaluators** Provides tools for to understand, validate and measure the capabilities of LLMs to solve planning problems

**Leonardo script** Contains specific scripts to launch and manage jobs on Leonardo HPC.

**models** Manages the communication with various LLMs, using "adapters" which creates a common interface for different backends like NVIDIA API.

**outputs** folder containing all the output from the LLMs, divided in 3 sub folders *raw*, *parsed*, *scored* to better separate respectively the raw output from the model, PDDL extracted and structured plans and validation and metrics results.

**prompts** contains text files each one used to explain to the LLM the rules of the domain and how to manage the problem in order to solve it.

**protocols** defines different strategies for test execution, for example direct plan (which is one shot) or iterative repair.

**reporting** contains scripts for creation of the plots used in the reports.

**runner** contains scripts useful to launch a single test or the whole suite.

**scripts** contains scripts for the computation of the numerical score assigned to each domain.

**slurm logs** Used to gather information for debug when the benchmark runs on Leonardo HPC

**tasks** PDDL problems and related instances selected for the runs, the instances are divided in 3 categories *easy*, *medium*, *hard*.

**tests** Used to run tests in preparation for the real run of the benchmark.

**utils** contains executable for the validator to be used during the run of benchmark framework

**advanced planning evaluation.py** it reads benchmark artifacts from the output folder and writes a model-centric JSON report in the folder results.

**clear outputs** is used to empty a folder during Leonardo tests

**run benchmark** is the file that runs the whole sequence of tests

The framework and its structure is organized in this way in order to allow the user to change fundamental parameters such as the numbers of iterations easily and effectively and then run the benchmark again with the modified parameters.

## 2.4 results

This folder contains all the plots and the output archive of all the analysis process; the outputs folder contains only raw data the results folder contains the analysis done on the data.

### 3 PDDL Domain and Problem Classification

Before running the benchmark, we classified the available PDDL material along two complementary axes. First, we assigned a numerical difficulty score to each domain and instance, so that benchmark results could be interpreted against an explicit complexity estimate. Second, we manually grouped the domains by the main type of reasoning they require.

The initial classification was computed on the 20 source domains stored in the archive of planning domains. The scoring code is located in the framework scripts folder, and the generated CSV/JSON reports are stored in the domain-complexity folder under analysis.

The script used for this step is `score_domains_complexity.py`.

#### 3.1 Difficulty Classification

##### 3.1.1 Instance raw scores

The complexity of a single problem instance is estimated through a weighted logarithmic combination of four structural metrics. The logarithm reduces the impact of very large feature values, preventing a single component from dominating the score.

The **raw score** is defined as:

$$S_{\text{raw}} = 0.45 \cdot \log(1 + A) + 0.20 \cdot \log(1 + G) + 0.20 \cdot \log(1 + N) + 0.15 \cdot T \quad (1)$$

where:

- **A (Actions):** estimated number of feasible grounded actions. This is computed by binding action parameters to objects compatible with the static preconditions observed in the initial state.
- **G (Goals):** total number of atomic goal conditions.
- **N (Numeric Score):** composite value representing numeric complexity:

$$N = E_{\text{num}} + 2 \cdot P_{\text{num}} + 3 \cdot G_{\text{num}} + M \quad (2)$$

where  $E_{\text{num}}$  is the number of numeric effects,  $P_{\text{num}}$  the number of numeric preconditions,  $G_{\text{num}}$  the number of numeric goal conditions, and  $M$  a binary flag indicating whether the problem defines a metric.

- **T (Topology Score):** spatial or relational complexity extracted from topology-like predicates such as *adj*, *connected*, or similar relations:

$$T = \log(1 + \text{Nodes}) + \log(1 + \text{Edges}) + \text{Density} \quad (3)$$

The weights are heuristic and can be changed depending on the aspect of complexity that should be emphasized. For example, increasing the weight of  $A$  would give more importance to the action branching factor.

The per-instance output is kept as a raw score. These raw scores are used to compare instances inside each domain and to compute domain-level summary statistics such as minimum, median, mean, and maximum raw score.

##### 3.1.2 Domain scores

The overall difficulty of a domain is not computed as a simple average of its instances. Instead, the framework uses the total amount of instance complexity available for that domain.

The **Domain Difficulty (0–100)** is computed by:

1. summing the  $S_{\text{raw}}$  values of all instances belonging to the domain, obtaining a *Total Raw Score*;
2. normalizing this total across all 20 source domains.

With this normalization, the easiest domain receives difficulty 0 and the hardest receives difficulty 100. In the current data, *fo-sailing* is the easiest domain, while *settlersnumeric* is the hardest one.

##### 3.1.3 Results obtained

The following heatmaps summarize the numerical classification. The first one shows the normalized domain difficulty score, while the second one compares raw-score statistics across domains.

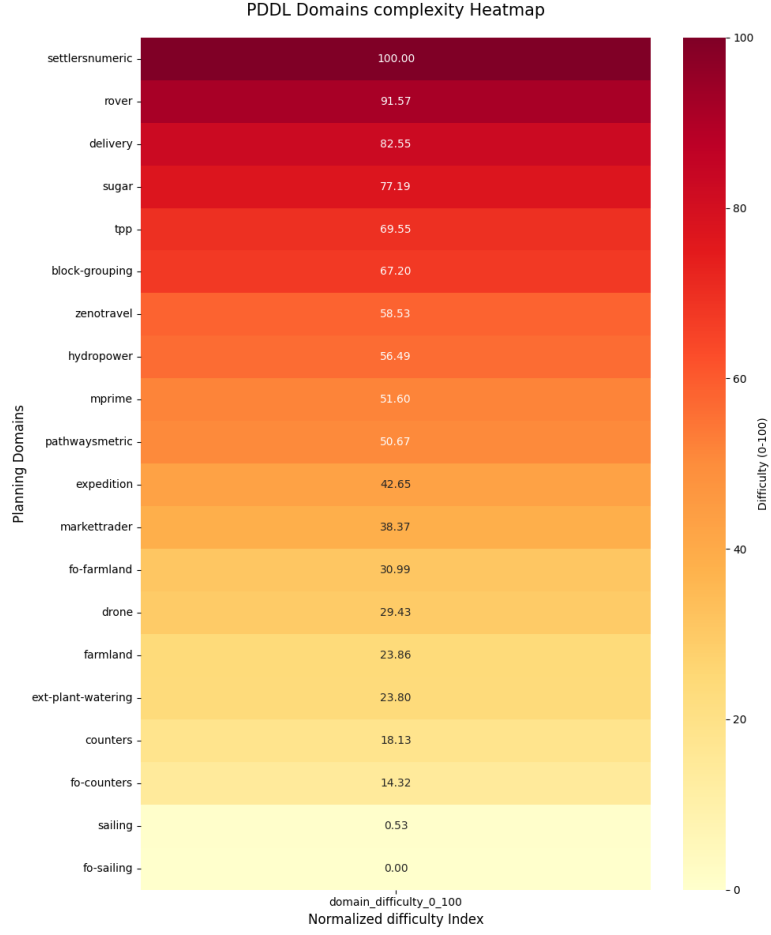


Figure 1: Normalized domain difficulty scores computed from total raw scores.



Figure 2: Raw-score statistics used to compare the source PDDL domains.

### 3.2 Reasoning Classification

In addition to numerical difficulty, we classified each domain according to the main type of reasoning required from the LLM. We used five categories:

- **Spatial:** problems involving objects, agents, or resources that must be moved or arranged in a spatial or

graph-like environment.

- **Numerical Optimization:** problems requiring management of numeric quantities, costs, resources, or objective functions.
- **Temporal Planning:** problems where the solution depends on ordering actions under time-related constraints or temporal relations.
- **Logistics:** problems focused on transportation, delivery, routing, or coordination between agents and goods.
- **Logic:** abstract symbolic problems where state transitions represent formal or mathematical transformations rather than physical actions.

Table 1 reports the manual classification of the 20 source domains. The rows are ordered from hardest to easiest according to the domain difficulty score described above. When a domain includes multiple reasoning aspects, we assign it to the category that dominates the problem structure. For example, *settlersnumeric* is classified as *Numerical Optimization* because its main challenge is the management of numeric resources and constraints.

| PDDL Problem       | Reasoning Category     |
|--------------------|------------------------|
| Settlersnumeric    | Numerical Optimization |
| Rover              | Logistic               |
| Delivery           | Logistic               |
| Sugar              | Numerical Optimization |
| Tpp                | Numerical Optimization |
| Block-Grouping     | Spatial                |
| Zenotrail          | Logistic               |
| Hydropower         | Temporal Planning      |
| Mprime             | Logic                  |
| Pathwaysmetric     | Numerical Optimization |
| Expedition         | Logistic               |
| Markettrader       | Numerical Optimization |
| Fo-farmland        | Numerical Optimization |
| Drone              | Spatial                |
| Farmland           | Numerical Optimization |
| Ext-Plant-Watering | Spatial                |
| Counters           | Numerical Optimization |
| Fo-Counters        | Numerical Optimization |
| Sailing            | Spatial                |
| Fo-Sailing         | Spatial                |

Table 1: Reasoning Category Classification Tab

### 3.3 Problems Chosen

To keep the benchmark compact while still covering different reasoning types, we selected two sets of three domains each. Both sets contain one domain from each of the three categories with enough alternatives for a hard/easy comparison: *Numerical Optimization*, *Spatial*, and *Logistics*. Temporal Planning and Logic were not selected because the source collection contains only one domain for each of those categories; this would not allow a meaningful difficulty comparison inside the same reasoning type.



The following table shows the domains available in the three selected categories, ordered by decreasing difficulty.

| Numerical Optimization | Spatial            | Logistics  |
|------------------------|--------------------|------------|
| Settlersnumeric        | Block-Grouping     | Rover      |
| Sugar                  | Drone              | Delivery   |
| Tpp                    | Ext-Plant-Watering | Zenotravel |
| Pathwaysmetric         | Sailing            | Expedition |
| Marketrader            | Fo-Sailing         |            |
| Fo-farmland            |                    |            |
| Farmland               |                    |            |
| Counters               |                    |            |
| Fo-Counters            |                    |            |

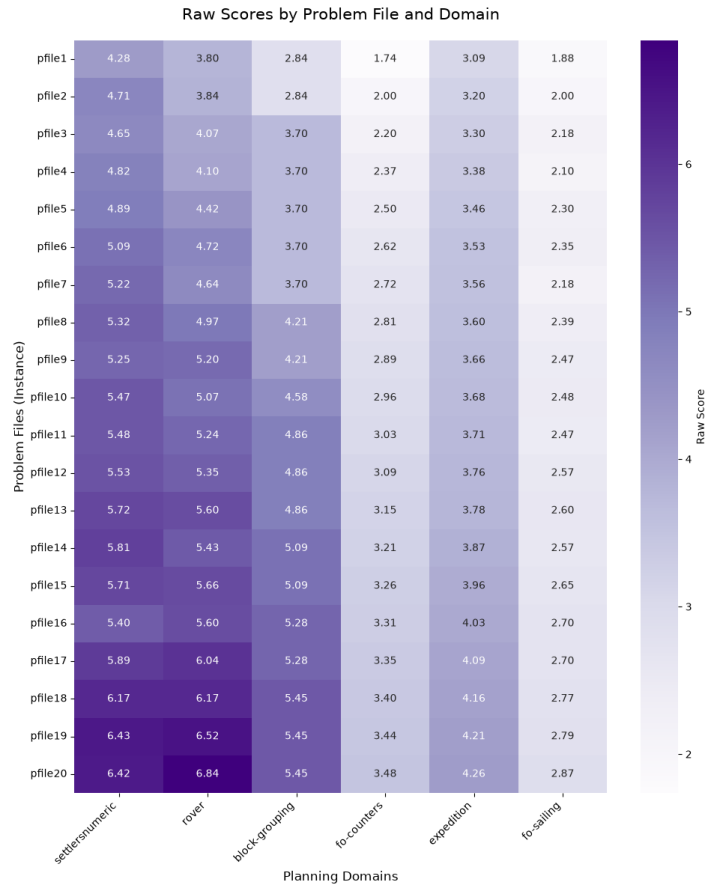
Table 2: Reasoning categories used for the final benchmark selection.

The final benchmark uses two domain sets:

- **Hard Set:** *settlersnumeric*, *block-grouping*, and *rover*.
- **Easy Set:** *fo-counters*, *fo-sailing*, and *expedition*.

The hard set contains the hardest domain in each selected reasoning category, while the easy set contains the easiest domain in the same categories. This keeps the reasoning-category distribution fixed and makes the difficulty gap easier to interpret when evaluating LLM outputs.

The selected domains still contain multiple instances. The following heatmap shows that, in general, pfiles increase in raw difficulty as the pfile index grows.



Based on this instance-level trend, we divided the selected pfiles into three difficulty tiers:

- **Easy:** pfile1–pfile4;
- **Medium:** pfile8–pfile11;
- **Hard:** pfile15–pfile18.

Therefore, the final benchmark does not use all available instances from each selected domain. Instead, it uses four instances per difficulty tier, for a total of 12 instances per selected domain.

## 4 Configured LLMs

After defining the PDDL problem set and the selected instance tiers, the next step was to define the model pool used to generate candidate plans. The benchmark framework was designed to support two execution backends: Leonardo HPC by CINECA and the NVIDIA API. Leonardo was used for local execution of Hugging Face models, while the NVIDIA API provided remote access to additional models without requiring every checkpoint to be installed and hosted locally.

### 4.1 Execution Backends

#### 4.1.1 Leonardo HPC

Leonardo HPC was the main platform considered for local model execution. The models were loaded through the Hugging Face backend, allowing direct control over the execution environment, GPU resources, numerical precision, prompt formatting, and generation parameters.

The selection of locally executable models was supported by whichLLM, a repository used to estimate which LLMs can run on a given hardware configuration and to compare models by release recency. In this project it was used as a preliminary feasibility reference, not as an automatic selection rule: the final local set was still constrained by the available Leonardo environment, dependency compatibility, and model access.

#### 4.1.2 NVIDIA API

The NVIDIA API was used as a remote inference backend. This reduced the setup effort required to test recent open-weight models and made it possible to include models that would have been less convenient to install or run locally.

The benchmark pipeline remained the same across both backends: prompts, parsing, validation, and scoring followed the same workflow, while only the model adapter changed. The NVIDIA API backend also introduced practical constraints, such as rate limits, temporary service availability, and model-specific API key requirements.

### 4.2 Selection Criteria

The models were selected according to four practical criteria:

- **Hardware feasibility:** local models had to be compatible with the available Leonardo environment, using whichLLM as a practical reference for hardware requirements.
- **Recency:** priority was given to recent LLM families, using whichLLM as a reference to compare model release dates.
- **Model diversity:** the benchmark included models from different families to avoid evaluating a single architectural lineage.
- **Evaluation availability:** models included in the reported comparison had to be executable through the same prompt, parsing, validation, and scoring workflow.

Using both backends made the framework more flexible than a Leonardo-only setup. Leonardo provided controlled local execution, while the NVIDIA API allowed the benchmark to cover additional hosted models through the same interface.

### 4.3 Configured Models

Table 3 reports the model pool considered by the benchmark framework. For the Hugging Face backend, the table lists the active local models in the Leonardo registry. For the NVIDIA API backend, it lists the remote models for which scored benchmark outputs were available in the collected runs. Some additional endpoints may remain present in the registry for future experiments, but they are not treated as part of the analyzed model set unless they have corresponding scored outputs.

| Backend      | Execution mode | Model identifier                  | Model name                 |
|--------------|----------------|-----------------------------------|----------------------------|
| Leonardo HPC | Local          | hf_gemma_4_31b_it                 | Gemma 4 31B IT             |
| Leonardo HPC | Local          | hf_phi_4                          | Phi-4                      |
| Leonardo HPC | Local          | hf_gpt_oss_120b                   | GPT-OSS 120B               |
| Leonardo HPC | Local          | hf_qwen_3_6_27b                   | Qwen 3.6 27B               |
| NVIDIA API   | Remote hosted  | nvidia_phi_4_mini_instruct        | Phi-4 Mini Instruct        |
| NVIDIA API   | Remote hosted  | nvidia_gpt_oss_120b               | GPT-OSS 120B               |
| NVIDIA API   | Remote hosted  | nvidia_kimi_k2_6                  | Kimi K2.6                  |
| NVIDIA API   | Remote hosted  | nvidia_llama_3_3_70b_instruct     | Llama 3.3 70B Instruct     |
| NVIDIA API   | Remote hosted  | nvidia_nemotron_3_nano_30b_a3b    | Nemotron 3 Nano 30B A3B    |
| NVIDIA API   | Remote hosted  | nvidia_nemotron_3_ultra_550b_a55b | Nemotron 3 Ultra 550B A55B |
| NVIDIA API   | Remote hosted  | nvidia_qwen_3_5_122b_a10b         | Qwen 3.5 122B A10B         |

Table 3: LLM models configured or included in the benchmark model pool.

## 5 Benchmark Output Parsing and Output Structure

The benchmark execution pipeline produces traceable and independently analyzable artifacts. All generated data is stored under `Benchmark_Framework/outputs`, which is divided according to the main stages of the evaluation: generation, parsing, and scoring.

This separation is a core design choice. It makes it possible to distinguish between different failure modes: a model may generate text that contains a reasonable plan but is not cleanly parseable, or it may produce a syntactically valid action sequence that still fails during formal validation. The artifacts are persisted when a benchmark case terminates, either because a valid plan has been found or because the configured maximum number of iterations has been reached.

The `outputs` directory is organized into three main subdirectories:

- **raw**: stores the direct model interaction and generation metadata.
- **parsed**: stores the structured action sequences extracted from the raw text.
- **scored**: stores validation outcomes, repair information, and aggregate metrics for the case.

Each layer follows the same hierarchical path:

`<run_id>/<model>/<protocol>/<domain>/<difficulty>/<instance>.json`

This layout makes the three artifacts for the same benchmark instance directly comparable across the raw, parsed, and scored layers.

### 5.1 Raw

The files in `outputs/raw/` are the closest record of the interaction with the LLM. Each artifact stores the prompts sent to the model, the model response, any provider-side reasoning text when available, and generation metadata such as token usage, latency, and notes.

This layer is the baseline record of what the model produced before any interpretation by the parser or validator.

#### 5.1.1 Raw File Structure

Listing 1 shows the main fields stored in a raw artifact. Provider-specific generation fields may vary across backends, but the surrounding structure is shared.

Listing 1: Schematic structure of a raw JSON artifact

```
{
  "model_id": "hf_gemma_4_31b_it",
  "task_id": "block-grouping_easy_pfile1",
  "protocol_id": "iterative_repair",
  "task_family": "block-grouping",
  "tier": "easy",
  "instance_id": "pfile1",
  "generation_time_seconds": 3526.41,
  "attempts": [
    {
      "iteration": 1,
      "generation_time_seconds": 623.25,

```

```

    "messages": [
      {
        "role": "system",
        "content": "You are a PDDL planning assistant..."
      },
      {
        "role": "user",
        "content": "TASK FAMILY: block-grouping..."
      }
    ],
    "generation": {
      "raw_text": "(move_block_left b1)\n(move_block_left b1)\n...",
      "reasoning_text": "Reasoning text, when available...",
      "usage": {
        "prompt_tokens": 2125,
        "completion_tokens": 4096,
        "total_tokens": 6221
      },
      "latency_s": 536.60,
      "notes": [],
      "generation_time_seconds": 623.25
    }
  }
]
}

```

## 5.2 Parsed

The `outputs/parsed/` directory contains the result of the first processing step after generation. The parser attempts to extract executable PDDL actions from the raw model text and stores the extracted plan in a structured form.

Each parsed artifact includes:

- the sequence of actions extracted from the visible model answer;
- parser diagnostics such as missing actions, unknown action names, wrong arity, markdown removal, or compressed-action expansion;
- a separate reasoning-derived action sequence when provider-side reasoning text contains a candidate plan;
- a pointer to the corresponding raw artifact.

This layer bridges the gap between unstructured natural language and the action sequence that can be submitted to validation. If no usable action can be extracted, the benchmark records a parser failure and the case can still be scored without invoking VAL on an empty plan.

### 5.2.1 Parsed File Structure

Listing 2 shows the current parsed artifact schema. The `raw` subsection represents the official plan extracted from the visible answer; the `reasoning` subsection is diagnostic and is used to study whether a plan was present only in reasoning text.

Listing 2: Schematic structure of a parsed JSON artifact

```

{
  "model_id": "hf_gemma_4_31b_it",
  "task_id": "block-grouping_easy_pfile1",
  "protocol_id": "iterative_repair",
  "task_family": "block-grouping",
  "tier": "easy",
  "instance_id": "pfile1",
  "generation_time_seconds": 3526.41,
  "raw_output_path": ".../outputs/raw/<run_id>/<model>/<protocol>/<domain>/<difficulty>/pfile1.json",
  "attempts": [
    {
      "iteration": 1,
      "generation_time_seconds": 623.25,
      "parsed_plan": {
        "raw": {
          "actions": [
            "(move_block_left b1)",
            "(move_block_left b1)"
          ],
          "format_issues": [
            "raw_text_contains_reasoning_like_content"
          ]
        }
      }
    }
  ]
}

```

```

    ],
    "contains_reasoning": true,
    "source_kind": "reasoning_like_raw"
  },
  "reasoning": {
    "actions": [
      "(move_block_left b1)",
      "(move_block_left b1)"
    ],
    "format_issues": [
      "multiple_reasoning_candidate_plans"
    ],
    "source_ref": {
      "artifact": "raw",
      "field": "generation.reasoning_text"
    }
  }
}
]
}

```

## 5.3 Scored

The final layer, `outputs/scored/`, contains the normalized outcome of the case. If the parser extracts at least one action, the action sequence is validated with VAL. If parsing fails, the scored artifact stores a normalized `parse_error` result instead. In both cases, the scored layer provides the fields used by the analysis notebooks and report metrics.

Each scored artifact includes:

- the final `solved` decision for the benchmark instance;
- the number of iterations used and whether the iteration limit was reached;
- aggregate metrics such as `validity_at_1`, `validity_at_k`, plan length, repair success, and failure type;
- one validation-style record for each attempt;
- repair feedback and reasoning-plan diagnostics when the iterative protocol or reasoning extraction uses them.

### 5.3.1 Scored File Structure

Listing 3 shows the main scored fields. The VAL-specific validation fields are nested under `validation_result`; prefix-validation, reasoning, and feedback fields belong to the attempt record itself.

Listing 3: Schematic structure of a scored JSON artifact

```

{
  "model_id": "hf_gemma_4_31b_it",
  "task_id": "block-grouping_easy_pfile1",
  "protocol_id": "iterative_repair",
  "task_family": "block-grouping",
  "tier": "easy",
  "instance_id": "pfile1",
  "solved": false,
  "iterations_used": 8,
  "max_iterations": 8,
  "stopped_by_iteration_limit": true,
  "generation_time_seconds": 3526.41,
  "metrics": {
    "validity_at_1": false,
    "validity_at_k": false,
    "repair_success": false,
    "iterations_to_valid": null,
    "plan_length": 30,
    "error_type": "unsatisfied_goal",
    "hit_iteration_limit": true
  },
  "raw_output_path": ".../outputs/raw/<run_id>/<model>/<protocol>/<domain>/<difficulty>/pfile1.json",
  "parsed_output_path": ".../outputs/parsed/<run_id>/<model>/<protocol>/<domain>/<difficulty>/pfile1.json",
  "attempts": [
    {
      "iteration": 1,
      "generation_time_seconds": 623.25,

```

```

    "validation_result": {
      "valid": false,
      "status": "invalid",
      "error_type": "unsatisfied_goal",
      "feedback_text": "The plan finishes without achieving the goal...",
      "failed_step": null,
      "failed_action": null,
      "goal_satisfied": false,
      "plan_length": 30,
      "validation_time_ms": 11,
      "raw_validator_output": "Checking plan...",
      "details": {
        "validator_kind": "external",
        "return_code": 1
      }
    },
    "first_valid_prefix_length": null,
    "first_valid_plan_text": null,
    "final_plan_valid": false,
    "extra_actions_after_first_valid": null,
    "reasoning_validation_result": null,
    "reasoning_first_valid_prefix_length": null,
    "reasoning_first_valid_plan_text": null,
    "reasoning_final_plan_valid": null,
    "reasoning_extra_actions_after_first_valid": null,
    "reasoning_candidate_count": null,
    "reasoning_valid_candidate_count": null,
    "reasoning_selected_candidate_index": null,
    "reasoning_selected_candidate_truncated": null,
    "feedback_to_next_iteration": "Use the validator feedback as a diagnostic hint..."
  }
]
}

```

## 6 Output Validation Using VAL

After the benchmark has obtained a textual answer from an LLM and extracted the candidate PDDL actions, the plan still has to be checked against the formal domain and problem definition. This is the role of VAL in the framework. The parser can say whether the model produced something that looks like an action sequence; VAL decides whether that sequence is actually executable and whether it reaches the goal.

### 6.1 Role of VAL in the Benchmark

VAL is used as an external symbolic validator. For each benchmark case, the framework provides VAL with three inputs:

- the PDDL domain file, containing predicates, actions, preconditions, and effects;
- the PDDL problem file, containing objects, initial state, and goal;
- the candidate plan extracted from the LLM output.

This separation is important because the benchmark does not rely on textual judgement to decide whether a plan is correct. A model can produce a plausible or well explained answer, but the answer is counted as solved only if the final action sequence validates against the corresponding PDDL task.

### 6.2 Validator Setup and Preflight

VAL is not a Python dependency of the project. It is an external executable that can be resolved in three ways: by putting `Validate` or `Validate.exe` on the system path, by passing an explicit executable path with `-validator-command`, or by using one of the bundled platform utilities under `utils/linux64/bin/` or `utils/win64/bin/`.

The benchmark can also run a task preflight before launching model generation. In this stage the framework checks the selected domain and problem files with VAL. This prevents malformed or unsupported PDDL inputs from being interpreted later as model failures. If VAL cannot be resolved, or if the preflight fails, the error is stored as an orchestration problem instead of being mixed with the LLM planning results.

### 6.3 Validation Flow

The validation path is the same for every model backend. The backend changes how the answer is generated, but not how the answer is checked:

LLM output  $\rightarrow$  parsed actions  $\rightarrow$  temporary plan file  $\rightarrow$  VAL(domain, problem, plan)  $\rightarrow$   
ValidationResult  $\rightarrow$  scored artifact

The temporary plan file is created only to invoke VAL with the expected command line interface. After validation, the framework normalizes the validator output into a common result schema. This makes the rest of the benchmark independent from the exact textual format printed by the external validator.

Only the validation summary belongs to the scored layer. The raw layer keeps the prompt and model generation payloads, the parsed layer keeps the extracted actions and parser diagnostics, and the scored layer keeps the validation results, repair data, and metrics. This lets the report discuss validation without repeating the full output structure described in the previous section.

### 6.4 Normalized Validation Result

The normalized validation record stores both the final decision and the diagnostic information needed for analysis and repair.

A parse failure and a validation failure are not the same event. If no clean PDDL action sequence can be extracted, VAL cannot judge the plan semantics and the attempt is treated as a formatting or parsing failure. If actions are extracted but VAL rejects them, the attempt reached the symbolic validation stage and the error describes a planning failure or a validator-side technical problem.

| Field                | Meaning                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| valid                | Boolean result used to decide whether the candidate plan solved the task.                                                                                 |
| status               | Normalized status such as <code>valid</code> , <code>invalid</code> , <code>parse_error</code> , <code>timeout</code> , or <code>validator_error</code> . |
| error_type           | Compact failure class used by metrics and repair feedback.                                                                                                |
| failed_step          | First failing step reported by VAL, when available.                                                                                                       |
| failed_action        | Problematic action extracted from the validator output, when available.                                                                                   |
| goal_satisfied       | Whether the final state satisfies the goal.                                                                                                               |
| plan_length          | Number of action lines submitted to validation.                                                                                                           |
| validation_time_ms   | Time spent in the external validator call.                                                                                                                |
| raw_validator_output | Original VAL output kept for debugging and later inspection.                                                                                              |

Table 4: Main fields stored in the normalized validation result.

### 6.5 Prefix Validation

The framework does not validate only the full action sequence. It validates every non-empty prefix of the parsed final answer. This provides a more precise view of the model behaviour.

For example, a model may reach the goal after five actions and then continue with unnecessary or invalid actions. In that case, validating only the complete sequence would mark the attempt as invalid, while prefix validation reveals that a valid solution was present inside the answer. The scored artifact therefore records the first valid prefix length, the first valid plan text, whether the final full plan is valid, and how many extra actions appear after the first valid prefix.

This distinction is useful for the later reasoning analysis: it separates models that never find a valid plan from models that find a plan but fail to stop cleanly after satisfying the goal.

### 6.6 Failure Types and Repair Feedback

VAL failures are mapped into a small shared taxonomy. The most relevant classes are:

- `invalid_precondition`: an action is not applicable in the current state;
- `unsatisfied_goal`: the sequence executes but does not reach the required goal;
- `unknown_action`: the plan contains an action not defined in the domain;
- `syntax_error`: the plan format is not accepted by the validator;

- `timeout`, `validator_unavailable`, and `validator_crash`: technical validation failures, kept separate from planning mistakes.

In the iterative repair protocol, this normalized failure information is turned into feedback for the next model attempt. The feedback can include the validation status, the error type, the failed step, the failed action, and a compact description of the validator output. This makes the repair loop grounded in symbolic validation rather than in a generic request to try again.

## 6.7 Feedback Generation for Iterative Repair

The repair feedback is generated as a deterministic diagnostic step over the previous attempt. It is not a generic request to try again: the feedback selects the most relevant failure signal and turns it into a constrained instruction for the next model call.

### 6.7.1 Inputs Used to Build Feedback

The feedback builder can use the previous final answer, the previous reasoning text when present, parser issues, the raw-plan validation result, the reasoning-plan validation result, prefix-validation fields, and alignment signals between the final answer and the reasoning trace. The protocol may also add a custom feedback prompt before the generated diagnostic text.

### 6.7.2 Decision Order

The feedback is written by following a priority order. Technical failures are handled first: if generation failed, VAL was unavailable, or validation timed out, the feedback states that the attempt could not yet be judged as a confirmed planning mistake and asks for a clean plan that can be validated again.

Formatting and parsing failures are handled next. If the final answer is empty, does not contain a valid action sequence, or mixes reasoning text into the final answer, the feedback focuses on output format rather than plan semantics: one parenthesized PDDL action per line, with no surrounding prose.

If the answer reached semantic validation, the feedback uses the normalized VAL fields. An `invalid_precondition` error points to action applicability or ordering. An `unsatisfied_goal` error points to a plan that executes but does not reach the goal. If VAL exposes a failed step or failed action, that information is included so the next attempt can repair the concrete failing part of the sequence.

After this, the feedback checks whether prefix validation or reasoning validation changes the diagnosis. If a valid prefix exists but the final answer continues with extra actions, the model is told to stop at the first valid solution. If the reasoning trace contains a valid plan but the final answer is invalid, the model is told to transfer the reasoning plan into the final answer. If reasoning and final answer are inconsistent, the model is asked to rebuild one coherent corrected sequence.

### 6.7.3 Feedback Templates

| Detected condition                    | Feedback content                                                                                               |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Technical validation failure          | Explain that the attempt could not be judged as a planning mistake and request a clean plan for re-validation. |
| Parser or format failure              | Ask for only one parenthesized PDDL action per line, without explanations or markdown.                         |
| Reasoning valid, final answer invalid | Ask to copy the valid reasoning plan into the final answer.                                                    |
| Valid prefix with extra actions       | Ask to stop immediately after the goal is reached.                                                             |
| <code>invalid_precondition</code>     | Ask to reorder or replace actions so every precondition is satisfied.                                          |
| <code>unsatisfied_goal</code>         | Ask to extend or modify the plan until the goal condition is reached.                                          |
| Low reasoning/final-answer alignment  | Ask to rebuild one coherent plan and make the final answer match it.                                           |

Table 5: How validation and reasoning diagnostics shape iterative-repair feedback.

### 6.7.4 Prompt-Budget Control and Final Constraint

The previous raw answer and reasoning text can be included in the feedback prompt, but they are truncated with separate limits. This keeps the useful evidence from the failed attempt while avoiding a prompt dominated by old outputs. After the diagnostic text is assembled, every branch ends with the same final constraint: the



next answer must contain only the complete PDDL action sequence, one parenthesized action per line, with no explanation, headings, markdown, numbering, or comments.

## 6.8 Reasoning Validation

Some models expose a separate reasoning trace in addition to the final answer. The benchmark keeps the final answer as the official plan used for scoring, but it can also validate an action sequence extracted from the reasoning text. This second validation is diagnostic: it helps identify cases where the model derives a valid plan in its reasoning but fails to transfer it correctly into the final answer.

The scored result is therefore intentionally conservative: the official solved flag comes from the final answer, not from a hidden or separate reasoning plan. Reasoning validation is preserved to explain model behaviour, not to replace the answer that would actually be submitted by the model.

This distinction avoids giving credit for hidden reasoning when the final answer is invalid, while still preserving useful information for the reasoning capability analysis in the following section.

## 6.9 Metrics Derived from Validation

The validation result is the source for the core execution metrics. A case is considered solved only when the normalized validation result is valid. From this record the framework derives `validity_at_1`, `validity_at_k`, `repair_success`, `iterations_to_valid`, `plan_length`, `error_type`, and whether the run hit the iteration limit.

Therefore, VAL is not only a final correctness check. It is the component that connects raw model generations to comparable benchmark scores.

## 6.10 Limitations

The validation layer deliberately checks formal plan correctness, not the quality of the natural-language reasoning. It also depends on the availability and behaviour of an external executable, so timeout and validator errors are reported separately from genuine planning failures. Finally, the parser that normalizes VAL output uses conservative textual heuristics; for this reason the raw validator output is stored together with the normalized fields whenever manual inspection is needed.

# 7 LLMs Reasoning Capabilities Evaluation

## 7.1 Pipeline and Data flow

This section maps the analysis script `advanced_planning_evaluation_sp.py` onto the benchmark artifacts. It explains how the three on-disk layers, `raw`, `parsed`, and `scored`, become one metric table and how that table feeds aggregate results and plots.

### 7.1.1 Pipeline Generic description

The analysis stage converts the three benchmark artifact layers into one canonical table, then derives metrics, aggregate tables and plots from that single source. The detailed implementation walk-through is preserved in Appendix A.1; this section keeps only the information needed to understand the evaluation flow.

| 3               | 1                   | 8                | 27               | 7                   |
|-----------------|---------------------|------------------|------------------|---------------------|
| artifact layers | canonical DataFrame | aggregate tables | registered plots | capability profiles |

### 7.1.2 The pipeline at a glance

The pipeline has a fixed data path:

`raw/` → `parsed/` → `scored/` → `df_metrics` → `aggregate tables` → `plots + report.json`

`raw/` stores the original model generations, `parsed/` stores the extracted action sequences, and `scored/` stores the validation outcomes produced after VAL has checked the candidate plan. The analysis script joins these layers with the selected PDDL task files, so every row can be evaluated against the exact domain and problem that generated it.

| Stage                | Purpose                                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Load artifacts       | Creates one row per tested instance, preserving model, domain, difficulty, protocol, validity and iteration information. |
| Index PDDL context   | Extracts legal actions, objects, initial state and numeric fluents from the task files.                                  |
| Compute row metrics  | Adds hallucination, executability, precondition-awareness and CoT-alignment metrics to each row.                         |
| Aggregate            | Builds model-level, domain-level and ranking tables from row-level metrics.                                              |
| Visualize and export | Generates the registered plots and serializes the report payload.                                                        |

Table 6: Compact view of the analysis pipeline.

### 7.1.3 The canonical table: `df_metrics`

`df_metrics` is the central object of the analysis. It starts from the artifact rows and is horizontally extended with computed columns. Its grain is one benchmark instance, not one retry attempt. This is important because success rates, hallucination rates and executability scores are averaged over comparable units.

The retry structure is still available: the loader preserves per-attempt lists in private columns, and only the metrics that need them, such as FASR, IWSR, iteration profiles and CoT alignment, read those lists.

| Input           | Role                                                                                         |
|-----------------|----------------------------------------------------------------------------------------------|
| Artifact rows   | Provide identity fields and direct outcomes.                                                 |
| Domain context  | Provides action schemas, legal action names, predicates and numeric fluents.                 |
| Problem context | Provides objects, initial propositions and numeric initial values for the specific instance. |
| Warnings        | Records missing or malformed inputs without dropping rows.                                   |

Table 7: Inputs used to build the canonical metric table.

### 7.1.4 PDDLSimulator role

Some metrics require more than the final VAL decision. A plan can be invalid because it fails immediately, because it follows legal actions in the wrong order, or because it invents a state that never existed. To separate these cases, the analysis uses a lightweight forward simulator over the parsed PDDL fragments.

The simulator starts from the problem initial state, applies each known action step by step, and stops at the first unsatisfied precondition. If a missing proposition was true before, the error is classified as sequencing; if it was never true, it is classified as state fabrication. VAL remains the official correctness oracle; the simulator is a diagnostic layer for explaining invalid plans.

## 7.2 Metrics scientific foundation

The metric set follows one central claim from the planning literature: binary validity is necessary, but not sufficient, to evaluate planning behaviour in LLMs. A model can fail because it does not know the vocabulary, cannot track state transitions, depends on repeated sampling, or produces chain-of-thought that does not match the emitted plan.

The extended paper-by-paper justification is preserved in Appendix A.2. In the body of the report, the relevant mapping is summarized in Table 8.

| Framework claim                                                                         | Supporting source                                      | Metric design consequence                                                          |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------|
| Validity alone hides different types of planning failure.                               | Valmeekam et al. 2022; Huang et al. 2022               | Report success rate together with execution and grounding metrics.                 |
| LLMs can exploit familiar action and object names instead of robust symbolic reasoning. | Valmeekam et al. 2023; Kambhampati 2024                | Measure action hallucination, object hallucination and inverse hallucination rate. |
| Retries can inflate success without proving genuine correction.                         | Stechly et al. 2023; Kambhampati 2024                  | Separate SR, FASR, IWSR and Retry Gap.                                             |
| Executable-looking plans can still fail because the model loses state coherence.        | Huang et al. 2022; Dziri et al. 2023; Vafa et al. 2024 | Use executability ratio, PAS and temporal distance.                                |
| Chain-of-thought may be post-hoc rather than causally connected to the answer.          | Lanham et al. 2023; Kambhampati 2024                   | Compare reasoning-derived plans with emitted plans through CoTaL.                  |
| Performance can vary strongly across domains and difficulty levels.                     | McCoy et al. 2024; Correa et al. 2025                  | Report domain-level results, within-domain rankings and cross-domain profiles.     |

Table 8: Compact scientific foundation of the metric set.

No single metric is treated as a final verdict. Success Rate answers whether a model eventually solved the task; FASR and IWSR ask how dependent that success was on retries; hallucination metrics test grounding; execution metrics inspect how far invalid plans progress; and CoTaL tests whether the reasoning trace is aligned with the final plan.

### 7.2.1 Literature-motivated metrics beyond the current set

| Candidate metric                | Source motivation                                    | What it would add                                                      |
|---------------------------------|------------------------------------------------------|------------------------------------------------------------------------|
| Goal-achievement partial credit | Valmeekam et al. 2022                                | Fraction of goal atoms achieved.                                       |
| Optimality ratio                | Classical planning benchmarks; Valmeekam et al. 2022 | Comparison between produced plan length and optimal plan length.       |
| Standard-obfuscated gap         | Valmeekam et al. 2023; Correa et al. 2025            | Direct measure of dependence on meaningful names.                      |
| World-model coherence score     | Vafa et al. 2024                                     | Test of whether different histories imply coherent future transitions. |

Table 9: Future metric extensions suggested by the literature.

## 7.3 Metric Families

### 7.3.1 Planning Efficiency

This family measures how often a model solves a task and how much that result depends on the repair loop. The distinction is essential: a model that solves on the first attempt is different from a model that reaches the same final success rate only after repeated retries.

| Metric    | Formula                                                           | What it measures                                      | Main interpretation                               |
|-----------|-------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------|
| SR        | $\text{mean}(\text{Valid})$                                       | Fraction of instances solved within the retry budget. | Headline outcome, but insufficient alone.         |
| FASR      | $\text{mean}(\text{Valid} \wedge \text{Iterations} = 1)$          | Fraction solved without feedback.                     | Strongest signal of zero-shot planning ability.   |
| IWSR      | $\text{mean}(1/k)$ for solved instances at iteration $k$ , else 0 | Success discounted by attempts needed.                | Rewards fast convergence and penalizes late wins. |
| Retry Gap | $\text{SR} - \text{FASR}$                                         | Share of success attributable to retries.             | Large gap means SR is retry-inflated.             |

Table 10: Planning-efficiency metrics.

The three success metrics obey  $\text{FASR} \leq \text{IWSR} \leq \text{SR}$ . If all three are close, the model succeeds early and the repair loop adds little. If SR is high but FASR is low, the model depends heavily on retries. The full implementation details are in Appendix A.3.

### 7.3.2 Execution and Reasoning

This family looks inside invalid plans. Instead of asking only whether VAL accepted the final sequence, it asks how far the plan executes and what kind of state-tracking error causes the first failure.

| Metric             | Formula                                                            | What it measures                                         | Main interpretation                                                  |
|--------------------|--------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------|
| Exec               | $\text{first\_failure}/\text{plan\_length}$                        | Prefix fraction that executes before the first failure.  | High value means the plan respects local transitions for many steps. |
| Sequencing errors  | Failed preconditions that were true earlier.                       | Whether the model had the needed state but misplaced it. | Ordering failure rather than total fabrication.                      |
| State fabrications | Failed preconditions that were never true.                         | Whether the model required an impossible state.          | Weak or incoherent world modelling.                                  |
| PAS                | $\frac{\text{sequencing}}{\text{sequencing} + \text{fabrication}}$ | Share of failures that are ordering mistakes.            | High PAS means failures are closer to valid causal reasoning.        |
| Temporal distance  | $\text{failure\_step} - \text{last\_true\_step}$                   | Distance from last valid prerequisite to failure.        | Large distance means long-horizon ordering collapse.                 |

Table 11: Execution and reasoning metrics derived from simulated replay.

These metrics should be read as a cascade. Executability is the first gate; PAS becomes meaningful only when the plan runs far enough for the simulator to observe state transitions. For near-solving models, very small failure counts make sequencing/fabrication proportions unstable, so the breakdown must be read with SR and FASR.

### 7.3.3 Grounding and Discriminatory

Grounding metrics test whether the model is using the formal vocabulary of the domain and problem. They are not success metrics by themselves: a model can use legal names and still fail. Their role is discriminatory, because they separate models that at least read the schema from models that invent actions or objects.

| Metric               | Formula                                                    | What it measures                                         | Main interpretation                                      |
|----------------------|------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|
| Action hallucination | $\frac{\text{illegal action names}}{\text{total actions}}$ | Emitted action names not present in the schema.          | High value means weak action grounding.                  |
| Fuzzy hallucination  | Strict hallucinations with edit distance $> 2$ .           | Genuine inventions versus spelling or formatting errors. | A strict/fuzzy gap suggests parser or formatting issues. |
| Object hallucination | $\frac{\text{illegal arguments}}{\text{total arguments}}$  | Arguments not present in problem objects.                | Catches legal verbs applied to invented entities.        |
| IHR                  | $1 - \text{action hallucination}$                          | Grounding score with higher values better.               | Used in the composite score and profiles.                |

Table 12: Grounding and discriminatory metrics.

The grounding layer is a gate for later interpretation. If a model hallucinates many action names, execution and CoT metrics become less reliable because there is no stable formal plan to inspect. Low hallucination only means the model is inside the legal vocabulary; it does not imply planning ability.

#### 7.3.4 CoT Alignment (CoTaL)

CoTaL measures whether the chain-of-thought is aligned with the final emitted plan. It is a diagnostic for reasoning faithfulness: high task success with low CoT alignment suggests that the reasoning text may be post-hoc, while high alignment on invalid plans suggests faithful reasoning that still fails during execution.

When both the final answer and the reasoning trace contain reconstructable plans, the strict score compares the two action sequences structurally:

$$\text{CoTaL} = 0.35 \cdot \text{LCS} + 0.25 \cdot \text{prefix} + 0.20 \cdot \text{bag overlap} \\ + 0.10 \cdot \text{length ratio} + 0.10 \cdot \text{repetition similarity}$$

| Status              | Data available                                      | How to read the score                                   |
|---------------------|-----------------------------------------------------|---------------------------------------------------------|
| Comparable plans    | Raw plan and reasoning plan both available.         | Structural CoTaL is meaningful and high confidence.     |
| Semantic proxy only | Raw plan and reasoning text, but no reasoning plan. | Low-confidence coverage signal, capped at 0.35.         |
| No reasoning text   | No usable reasoning signal.                         | CoTaL is unavailable and should not be treated as zero. |

Table 13: Compact CoTaL status interpretation.

The main caveat is coverage. In sparse runs most attempts may not expose a separate reasoning plan, so a high average CoTaL can be based on very few instances. For this reason CoTaL should always be reported with its status mix or sample count.

#### 7.3.5 PS (Composite Planning Score)

The Composite Planning Score (PS) gives a compact leaderboard, but it is not a replacement for the diagnostic metrics. It combines zero-shot success, retry efficiency, execution quality, grounding and precondition awareness into one bounded score so that models can be sorted quickly before inspecting their component profiles.

$$\text{base} = 0.25 \cdot \text{FASR} + 0.20 \cdot \text{IWSR} + 0.20 \cdot \text{Exec} \\ + 0.20 \cdot (1 - \text{Hallucination}) + 0.15 \cdot \text{PAS}$$

When a strict CoT-alignment score is available, a small optional bonus is added and the result is renormalized so PS remains in  $[0, 1]$ .

| Component         | Weight | Reason                                                                           |
|-------------------|--------|----------------------------------------------------------------------------------|
| FASR              | 0.25   | Highest weight because first-attempt success is the least retry-inflated signal. |
| IWSR              | 0.20   | Rewards fast convergence and penalizes late retry success.                       |
| Exec              | 0.20   | Measures structural plan quality even when the goal is not reached.              |
| 1 - Hallucination | 0.20   | Captures grounding in the legal action vocabulary.                               |
| PAS               | 0.15   | Distinguishes sequencing errors from state fabrications.                         |
| CoT bonus         | +0.05  | Rewards faithful reasoning only when strict CoT alignment exists.                |

Table 14: Composite Planning Score components.

PS should always be reported with the component table or radar view: the same score can come from very different behaviours, such as strong zero-shot success with weak execution diagnostics or balanced medium performance across all axes.

## 7.4 Cross-Domain and Cross-LLM comparison

Within-domain ranking and its direction rules, the per-domain heatmap normalisation caveat, rank variance (generalist vs specialist), Spearman domain correlation, the failure-mode taxonomy quadrants, and the seven literature-grounded capability profiles with their NaN-aware assignment logic.

A single overall leaderboard hides the most scientifically interesting fact about a model: *where* it is good. This view documents the three machinery pieces that turn per-domain metrics into comparative claims — within-domain ranking, rank variance, and the Spearman domain-correlation matrix — and then the failure-mode taxonomy and the seven literature-grounded capability profiles that convert those numbers into a named diagnosis.

### 7.4.1 Within-domain ranking: why rank, not raw value

Absolute metric values are not comparable across domains — a 0.30 success rate in a hard transport domain and a 0.30 in an easy gripper domain mean different things. The framework therefore ranks models **inside each domain separately**, on every metric in `RANK_METRICS`, then works with ranks. Ranking removes the domain-difficulty offset and answers the only question that survives it: *among the models that faced this exact domain, who placed where?*

The direction of "good" is metric-specific and is encoded once in `RANK_METRICS`. Maximise for Success\_Rate, FASR, IWSR, Exec, PAS, CoT\_Alignment; minimise for Hallucination, Retry\_Gap, Temporal\_Distance. The rank call makes the direction explicit:

```
for metric, direction in RANK_METRICS.items():
    rank_table[metric] = sub[metric].rank(
        ascending = direction == "min",      # min-metrics: smallest value = rank 1
        method     = "min",                  # ties share the best rank
        na_option  = "bottom",               # missing values rank last, never first
    )
# rank 1 = best in this domain on this metric
```

#### The rationale of the procedure

Ranking is a deliberate refusal to average raw scores across heterogeneous domains. "Model A beats Model B" is only ever asserted

*within a domain on a named metric*

; a global statement is then built by aggregating those local verdicts (mean rank, rank variance), never by averaging raw metric values that live on incomparable scales.

`na_option="bottom"`

is a fairness rule: a model that produced no usable plan cannot be handed a good rank by accident.

### 7.4.2 The ranking heatmap and its hard caveat

The `domain_ranking_heatmap` renders one sub-grid per domain (rows = models, columns = `RANK_METRICS`). To share one colour scale across domains with different model counts, each raw rank is min-max normalised inside its domain to `[0,1]`:

$$\text{norm\_rank} = \frac{\text{rank} - 1}{\max(\text{n\_models} - 1, 1)}$$

$\in [0, 1]$ , 0 = best in domain, 1 = worst in domain  
Colormap: RdYlGn\_r, green = best.

#### Do not read normalised cells as cross-domain quality

The normalisation is

##### per-domain

. A 0.00 (green) cell means "best

*among the models present in this domain*

", not "good in absolute terms". A model can be deep green in a domain where every model failed.

Compare cells

*down a column within one domain block*

; never compare the same model's colour across two different domain blocks as if it were a score. The colour encodes local standing, not capability.

### 7.4.3 Rank variance: generalist vs specialist

Rank variance compresses "where is this model good" into one number. For each model it takes the standard deviation of its within-domain rank across domains, per metric, then averages those across metrics (`mean_rank_std`). A model that places ~ 3rd everywhere has near-zero variance (a generalist); a model that wins one domain and trails in another has high variance (a domain specialist).

```
var_df = rank_within_domain.groupby("Model")[RANK_METRICS_cols].std()
var_df["mean_rank_std"] = var_df[RANK_METRICS_cols].mean(axis=1)
var_df = var_df.sort_values("mean_rank_std", ascending=False) # specialists on top
```

`phi_4` carries the highest `mean_rank_std`  $\approx 2.394$  (most domain-specific — strong in a domain or two, weak elsewhere) while `gpt_oss` sits lowest at  $\approx 1.349$  (most consistent across domains). High variance is not "bad": it flags a model whose single overall number is least trustworthy, because its behaviour is domain-contingent.

#### Spec says "normalised rank", code uses raw rank

The plot docstring describes rank variance as the std of

*normalised*

rank, but the implementation computes std over the

*raw*

integer ranks in

`rank_within_domain`

(the normalisation is applied only inside the heatmap). With a fixed model count the two differ by a constant factor, so the ordering of models is unaffected — but the absolute

`mean_rank_std`

values are on the raw-rank scale, not [0,1]. Worth aligning if you ever compare variance magnitudes across runs with different model counts.

### 7.4.4 Domain correlation: redundancy vs complementarity

The `domain_correlation` plot is a model×model Spearman  $\rho$  matrix built from each model's vector of per-domain success rates. Two models with  $\rho \approx 1$  succeed and fail on the same domains — they are *redundant* probes of capability. Two with low or negative  $\rho$  stress different things — they are *complementary*, and a benchmark wants both. It is a tool for curating the model set and the domain set, not for ranking.

```
sr_pivot = df_metrics.pivot_table(index="Model", columns="Domain", values="Success_Rate")
corr = sr_pivot.T.corr(method="spearman") # Model \(\times\) Model rank correlation
```

#### Correlation needs a domain spread

Spearman over success-rate vectors is only meaningful with several domains and some variance in them. In the bundled run the domain count is small and many models sit at  $\text{SR} \approx 0$ , so  $\rho$  is dominated by ties

and should be read as illustrative of the method, not as a stable finding.

#### 7.4.5 Breadth of competence: stacked PS and the SR heatmap

`ps_by_domain_stacked` stacks each model's per-domain Planning Score contributions into one bar. Two models can reach the same total bar height by opposite routes: a tall single segment (high PS concentrated in one domain) versus many even segments (competence spread across domains). The stack makes "broad vs concentrated" visible at a glance — exactly the distinction the overall PS scalar destroys.

The `success_rate_heatmap` is the raw companion: model×domain SR with no normalisation, so here — unlike the rank heatmap — cells *are* comparable across the whole grid. Use it to spot empty columns (domains no model solved → an infrastructure or difficulty problem, not a model problem) and empty rows (models that solved nothing).

#### 7.4.6 The failure-mode taxonomy plot

This is the framework's single most diagnostic figure. It places every model on two orthogonal failure axes — **hallucination rate**(x, grounding failure) against **PAS**(y, precondition awareness) — encoding **FASR as bubble size** and **SR as colour**. The point is that "the model failed" is not one phenomenon: the two axes separate four qualitatively different failure regimes.

| Quadrant               | Reading                                           | Diagnosis                       |
|------------------------|---------------------------------------------------|---------------------------------|
| low halluc · high PAS  | grounded, legal names, but mis-orders actions     | Sequencing failure (e.g. gemma) |
| low halluc · low PAS   | legal names, no coherent state model              | Vocabulary-only                 |
| high halluc · high PAS | invents actions yet shows some precondition sense | Confused / mixed                |
| high halluc · low PAS  | not reading the domain; everything ungrounded     | No grounding                    |

##### What the taxonomy lets you conclude

Two models with identical success rates can sit in different quadrants — and that difference is the actual research finding. In the bundled run gemma is the only sequencing-heavy model (failure breakdown 20 sequencing vs 28 fabrication, seq-proportion 0.417), so it lives in the grounded/high-PAS corner while the flat-zero models collapse toward the ungrounded corner. The taxonomy turns an undifferentiated pile of failures into named, citable failure phases.

#### 7.4.7 The seven capability profiles

The profiles are the taxonomy made formal: each is a conjunction of threshold conditions over  $\{SR, FASR, IWSR, RG, IHR, ER, PAS, CoT, p_{valid,k,flat}\}$ , and each maps to a specific claim in the planning literature. They are the bridge from "here are nine numbers" to "this model is doing X".

#### 7.4.8 How a model is assigned a profile

Assignment is deliberately conservative. Every profile is evaluated; the **first exact match**(all required conditions met, none unavailable) wins. If nothing matches exactly the model is labelled `Mixed/Unclassified` and reported against its *closest* profile — the one with the highest matched-condition count, ties broken by fewest required conditions.

```

evaluations = [evaluate_profile_conditions(metrics, p) for p in PROFILE_DEFINITIONS]
exact = next((e for e in evaluations if e["exact"]), None)
if exact is None:
    best = max(evaluations, key=lambda e: (e["match_count"], -e["required_count"]))
    assigned = "Mixed/Unclassified" # honest about partial matches
else:
    assigned = exact["profile"]

```



| Profile                                                 | Conditions                                                                                        | Interpretation · reference                                        |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Genuine Planner</b>                                  | $SR > 0.70 \cdot FASR > 0.50 \cdot RG < 0.10 \cdot IHR > 0.90 \cdot CoT > 0.70$                   | Reads the schema, solves first try, reasoning tracks the plan.    |
| Strongest claim                                         |                                                                                                   |                                                                   |
| <b>Efficient Corrector</b>                              | $SR > 0.50 \cdot FASR < 0.30 \cdot  SR - IWSR  \leq 0.10 \cdot 0.10 \leq RG \leq 0.30$            | Rarely first-try, but repairs land early so IWSR stays near SR.   |
| Stechly 2023 (verify via $P(\text{Valid} \mathbf{k})$ ) |                                                                                                   |                                                                   |
| <b>Stochastic Searcher</b>                              | $SR \geq 0.30 \cdot FASR < 0.20 \cdot RG > 0.30 \cdot [\text{opt}]$<br>$P(\text{Valid}   k)$ flat | Success inflated by retries; attempts behave like resampling.     |
| Stechly 2023                                            |                                                                                                   |                                                                   |
| <b>Lucky Retriever</b>                                  | $SR > 0.60 \cdot FASR > 0.50 \cdot IHR < 0.30 \cdot CoT < 0.40$                                   | Solves from surface patterns; weak grounding, post-hoc reasoning. |
| Kambhampati 2024 · Valmeekam obfuscation                |                                                                                                   |                                                                   |
| <b>Understander</b>                                     | $SR < 0.20 \cdot FASR < 0.10 \cdot ER > 0.70 \cdot IHR > 0.90 \cdot PAS > 0.70$                   | Knows vocabulary & local preconditions, fails long-horizon order. |
| Valmeekam Phase 2 · Dziri 2023                          |                                                                                                   |                                                                   |
| <b>Vocabulary-Only</b>                                  | $SR < 0.15 \cdot IHR > 0.90 \cdot 0.30 \leq ER \leq 0.70 \cdot PAS < 0.30$                        | Legal action names, no coherent state-transition model.           |
| Kambhampati 2024 · Vafa 2024                            |                                                                                                   |                                                                   |
| <b>No Grounding</b>                                     | $SR \leq 0.05 \cdot IHR < 0.60 \cdot ER < 0.30 \cdot PAS < 0.30$                                  | Not reading the formal domain at all; everything ungrounded.      |
| Valmeekam 2023 · McCoy 2024                             |                                                                                                   |                                                                   |

#### NaN-aware conditions: "unavailable" $\neq$ "missing"

A condition whose metric is NaN (e.g. CoT when reasoning was never produced) is marked *unavailable*

, not  
*failed*

. This is what stops a model being thrown out of "Genuine Planner" merely because CoT could not be computed — and it is why the report exposes the full matched/missing/unavailable breakdown per profile, so you can see exactly how close a model sat to each boundary rather than trusting a bare label.

#### Profiles are thresholds, not ground truth

The cut-points ( $SR > 0.70$ ,  $IHR > 0.90$ , ...) encode the literature's qualitative claims as hard numbers; they are defensible but not sacred. A model just under a boundary is reported as Mixed against its closest profile precisely so the threshold choice stays visible and auditable. Treat the label as a hypothesis the condition breakdown lets you check, not a verdict.

### 7.4.9 Reading the metric set together: the diagnostic framework

No single metric is a verdict. The scientific meaning lives in *combinations*— and the within-domain ranking heatmap is where they are read side by side. There are three reading directions, and then a small set of metric pairs that must never be read in isolation.

#### Three ways to read the ranking heatmap.

| Direction                    | What it tells you                                                                                                                                                                       |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A <b>row</b> (one model)     | Scan all metrics: uniformly green = leads on every dimension here; uniformly red = worst on everything; a mixed row is a capability profile — strengths and weaknesses at once.         |
| A <b>column</b> (one metric) | Which model dominates that metric. A column all one colour means the metric doesn't discriminate in this domain.                                                                        |
| <b>Cell blocks</b>           | Green SR+FASR but red halluc+CoT = succeeding despite poor grounding (pattern-matching). Red SR but green exec+PAS = failing at the plan level yet showing process-level understanding. |

**The metric pairs to always read together. Capability triangle — SR + FASR + IWSR + RG**(is performance genuine or scaffolded by retriesc). Read SR and FASR first, then RG = SR - FASR, then IWSR position.

| SR     | FASR   | RG               | IWSR                  | Diagnosis                                                     |
|--------|--------|------------------|-----------------------|---------------------------------------------------------------|
| High   | High   | $\approx 0-0.10$ | $\approx \text{FASR}$ | Genuine zero-shot planner — retries irrelevant.               |
| High   | Medium | $0.10-0.25$      | $\approx \text{SR}$   | Efficient corrector — converges on attempts 2–3.              |
| High   | Low    | $> 0.30$         | $\approx \text{FASR}$ | Stochastic search — SR inflated; late retries are re-samples. |
| Medium | Low    | $0.20-0.40$      | mid                   | Partial capability, retry-scaffolded.                         |
| Low    | Low    | $\approx 0-0.05$ | $\approx \text{FASR}$ | No capability — nothing for retries to inflate.               |

**Failure anatomy — ER + PAS**(most useful when SR < 0.15: how deep does it run, and why did it stopc)

| ER     | PAS    | What the model is doing                                                            | Valmeekam phase        |
|--------|--------|------------------------------------------------------------------------------------|------------------------|
| > 0.70 | > 0.70 | Runs deep, ordering errors near the goal — understands domain, mis-sequences late. | Phase 2 (long-horizon) |
| > 0.70 | < 0.30 | Runs deep but fabricates states — knows vocabulary, invents transitions.           | Phase 1/2 boundary     |
| < 0.30 | > 0.70 | Fails early, but on ordering of the few real steps. Rare.                          | Phase 2 + vocab issues |
| < 0.30 | < 0.30 | Fails immediately, invents states from step 1.                                     | Phase 1 (vocabulary)   |
| Medium | NaN    | Mixed; PAS undefined — check halluc gate and temporal distance.                    | Mixed                  |

**Reasoning test — SR + CoT and refinement pair — IHR + PAS**(genuine vs retrieved success; and which kind of failure).

#### IHR is a gate, not just a metric

If inverse-hallucination rate falls below  $\sim 0.6$ , ER, PAS and CoTaL are all unreliable: the model is failing at vocabulary, so the downstream causal metrics have nothing trustworthy to measure. Always clear the IHR gate before interpreting the process metrics.

**The diagnostic tree.** The pairs above compose into a single ordered procedure — read the metrics in this sequence, taking the branch that matches, and the model classifies itself.

## 7.5 Graph Catalogue

Every figure the pipeline emits, documented by scope, reading procedure, the data-availability problems that distort it on sparse runs, and the justification for its chart type — grouped by metric family.

The pipeline emits up to 27 figures, registered by name in `PLOT_DESCRIPTIONS` and written by `maybe_make_plots`. This catalogue documents each one along the four axes that decide whether a figure earns its place: its **scope**(what slice of data it shows), **how to read it**, the **data-availability problems** that distort it on sparse runs, and the **justification** for that chart type over the alternatives.

#### One reading rule that governs half the figures

Several plots are bounded  $[0,1]$  and zero-inflated: on weak models the bars genuinely collapse to zero, and a zero can mean either "good" (no hallucination) or "nothing produced" (no actions emitted). Whenever a bar is flat-zero, disambiguate against Success\_Rate and plan counts before reading it as a strength. The

| Pair                    | Signature                                                                         | Meaning                                                                    |
|-------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| SR high · CoT > 0.70    | —                                                                                 | Reasoning guides grounded plans — strongest evidence of planning (Lanham). |
| SR high · CoT < 0.40    | —                                                                                 | Fake reasoning — correct plans, post-hoc CoT (Kambhampati).                |
| SR low · CoT > 0.70     | —                                                                                 | Faithful but failing — reasoning grounded, execution fails.                |
| IHR > 0.90 · PAS > 0.70 | top-left (good)                                                                   | Knows vocabulary, ordering errors only — most fixable.                     |
| IHR > 0.90 · PAS < 0.30 | bottom-left                                                                       | Knows words, lacks world model.                                            |
| IHR < 0.60 · PAS low    | bottom-right (worst)                                                              | Near-random output — no grounding, no causal model.                        |
| IHR < 0.60              | STOP. Not reading the schema — vocabulary failure. Do not interpret exec/PAS/CoT. |                                                                            |
| IHR 0.60–0.90           | Proceed with caution; note partial vocabulary issues.                             |                                                                            |
| IHR > 0.90              | Proceed to Level 1.                                                               |                                                                            |

atlas notes this per-plot where it bites.

### 7.5.1 Grounding / discriminatory

#### Hallucination heatmap

`hallucination_heatmap`

**chart** heatmap

**scope** Mean action-hallucination rate per model×domain. Lower = better grounded.

**how to read** Read down a column to compare models in one domain; dark = invents illegal actions. Empty cells = no plans produced there.

**data problems** A model with  $SR \approx 0$  can still show 0 hallucination if it never emitted a parseable action — read alongside Exec.

**why this form** Heatmap is right because the data is a dense model×domain grid of one bounded scalar; small multiples would waste space.

#### Action hallucination (bars)

`hallucination_by_model_domain`

**chart** grouped bars

**scope** Same action-hallucination rate, grouped bars per model across domains.

**how to read** Bar height = fraction of emitted actions whose name is not in the domain schema.

**data problems** Zero-height bars are ambiguous: genuinely grounded vs nothing emitted. Cross-check plan counts.

**why this form** Bars expose per-domain spread for one model better than a heatmap row when domain count is small.

#### Object hallucination

`object_hallucination_by_model_domain`

**chart** grouped bars

**scope** Fraction of action arguments referencing objects absent from the problem file.

**how to read** Separates 'wrong object' errors from 'wrong action' errors; compare against the action-hallucination twin.

**data problems** Undefined when a model emits no arguments; appears as 0 and must not be read as 'good'.

**why this form** Splitting object- from action-hallucination is the whole point — two distinct grounding failures need two plots.

|              |                                                                                   |
|--------------|-----------------------------------------------------------------------------------|
| SR < 0.15    | Almost no valid plans — skip to Level 4; the question is which failure dominates. |
| SR 0.15–0.50 | Partial capability — Level 2.                                                     |
| SR > 0.50    | Can plan here — Level 2 to test authenticity.                                     |

---

|              |                                                                         |
|--------------|-------------------------------------------------------------------------|
| RG < 0.10    | Retry-independent (FASR $\approx$ SR) — Level 3 + Level 5.              |
| RG 0.10–0.30 | Moderate dependency — report SR and FASR; Level 3.                      |
| RG > 0.30    | Strong dependency — SR inflated, use FASR; confirm via flat P(Valid k). |

### 7.5.2 Execution & reasoning

#### Executability ratio

`executability_by_model_domain`

**chart** box / strip

**scope** Distribution of `exec_ratio` (valid-prefix length / plan length) per model $\times$ domain.

**how to read** Higher = plans run further before the first precondition break. 1.0 = fully executable (often also valid).

**data problems** Single-instance domains give a degenerate ‘distribution’ of one point; spread is only meaningful with several instances.

**why this form** A distribution view (not a mean) is justified because `exec_ratio` varies instance-to-instance and the spread is informative.

#### Executability vs plan length

`executability_vs_length`

**chart** scatter (faceted)

**scope** `exec_ratio` against plan length, one facet per domain.

**how to read** A downward trend = the model breaks down on longer horizons (the Embers-of-Autoregression length cliff).

**data problems** Needs many instances spanning a length range; sparse runs show a near-empty scatter with no visible trend.

**why this form** Scatter is the only honest way to show a per-instance relationship between two continuous quantities.

#### Failure type breakdown

`failure_type_breakdown`

**chart** stacked bars

**scope** Per model: count of sequencing errors vs state fabrications among failed plans.

**how to read** Tall sequencing segment = grounded model failing on order; tall fabrication segment = invents unsatisfiable states.

**data problems** Models with almost no failures (or no plans) give tiny/empty bars — proportion is noisy at low counts.

**why this form** Stacking the two mutually exclusive failure causes shows both the mix and the absolute volume in one bar.

#### Temporal distance

`temporal_distance_by_model`

**chart** bars

**scope** Mean gap (steps) between the last true-precondition step and the actual failure step, sequencing errors only.

**how to read** Larger = the model stays internally consistent for longer before diverging (a ‘compositional reach’ proxy).

**data problems** Defined only for sequencing errors; models with none get NaN/absent bars — absence is not zero.

**why this form** A simple per-model bar suffices because temporal distance is already a per-model mean of a single quantity.

|                                     |                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------------|
| IWSR $\approx$ FASR                 | All success on attempt 1 — confirm with declining $P(\text{Valid} \mathbf{k})$ .       |
| IWSR near SR                        | Efficient corrector (attempts 2–3) — verify with rising $P(\text{Valid} \mathbf{k})$ . |
| IWSR near FASR, high RG             | Late-retry luck (attempts 4–5) — stochastic; flat $P(\text{Valid} \mathbf{k})$ .       |
| ER $> 0.70 \cdot \text{PAS} > 0.70$ | Long-horizon sequencing failure — Phase 2 (Valmeekam).                                 |
| ER $> 0.70 \cdot \text{PAS} < 0.30$ | Runs deep but fabricates — world-model incoherence (Vafa).                             |
| ER $< 0.30 \cdot \text{PAS} < 0.30$ | Fails immediately — Phase 1 fabrication.                                               |
| ER 0.30–0.70                        | Mixed — check temporal distance (low = near-miss, high = broken causal model).         |

### 7.5.3 Planning efficiency

#### FASR by model $\times$ domain

`fasr_by_model_domain`

**chart** grouped bars

**scope** First-attempt success rate per model across domains — success with no repair.

**how to read** The retry-free skill signal; compare against SR bars to see how much each model leans on retries.

**data problems** Bounded [0,1]; zero bars common and meaningful (no first-try solves).

**why this form** Bars per domain make the zero-inflation and the few non-zero domains immediately legible.

#### SR vs FASR

`sr_vs_fasr`

**chart** paired bars

**scope** Overall success rate beside first-attempt success rate, per model.

**how to read** The vertical gap between the pair is the Retry Gap; a large gap = retry-dependent success.

**data problems** Both can be 0 for weak models — the pair collapses and the gap is trivially zero.

**why this form** Pairing the two bars makes the gap a visual quantity instead of a number to subtract mentally.

#### FASR by difficulty

`fasr_by_difficulty`

**chart** bars

**scope** First-attempt success rate aggregated by difficulty tier.

**how to read** A monotone decline across tiers is the expected difficulty gradient; a cliff signals a capability boundary.

**data problems** Requires a populated difficulty field; if tiers are unlabelled or unbalanced the bars mislead.

**why this form** Binning by difficulty is the natural way to expose the McCoy difficulty cliff.

#### IWSR by model $\times$ domain

`iwsr_by_model_domain`

**chart** grouped bars

**scope** Iteration-weighted success rate (1/iterations on success, else 0) per model $\times$ domain.

**how to read** Discounts late successes; sits between FASR and SR. Closeness to SR = efficient repair.

**data problems** Bounded [0,1]; like FASR it is zero-inflated for weak models.

**why this form** Same grid logic as FASR; bars keep the three efficiency metrics visually comparable.

#### SR · FASR · IWSR triptych

`sr_fasr_iwsr_by_model`

**chart** grouped bars

**scope** All three success metrics side by side per model.

**how to read** Read the descent SR $\rightarrow$ IWSR $\rightarrow$ FASR: a steep drop = retry-inflated; a flat triple = genuine first-try skill.

|                      |                                                                          |
|----------------------|--------------------------------------------------------------------------|
| CoT > 0.70 · SR high | Genuine deliberation — strongest result.                                 |
| CoT < 0.40 · SR high | Fake reasoning — post-hoc CoT (Kambhampati); report as caveat.           |
| CoT > 0.70 · SR low  | Faithful but failing — likely long-horizon (if ER also high).            |
| CoT < 0.40 · SR low  | Complete grounding failure — return to Level 0 for a masked vocab issue. |

**data problems** For SR≈0 models all three vanish together, conveying little beyond 'failed'.  
**why this form** Co-plotting the three is the cleanest single picture of retry dependence per model.

#### Retry gap

`retry_gap_by_model`

**chart** bars

**scope** SR - FASR per model; the Stechly retry-dependence signal.

**how to read** Higher = more of the reported success came only after retries.

**data problems** Zero gap is ambiguous: either pure first-try skill or total failure (SR=FASR=0). Disambiguate with SR.

**why this form** A single derived bar per model is appropriate; the gap is one number with a clear sign.

#### FASR vs IWSR scatter

`fasr_iwsr_scatter`

**chart** bubble scatter

**scope** FASR (x) vs IWSR (y) with SR encoded as dot size and colour.

**how to read** On-diagonal = first-try-dominated; below-diagonal with large dot = success built by efficient repair.

**data problems** Overplotting at the origin when many models score ~ 0; jitter or label to separate.

**why this form** Encoding three metrics (x,y,size/colour) in one scatter is denser than three separate bar charts.

#### P(Valid | reached k)

`p_valid_given_k`

**chart** line

**scope** Probability a plan is valid given the model reached attempt k, across k.

**how to read** Rising curve = Efficient Corrector (later attempts really are better); flat line = Stochastic Searcher (resampling).

**data problems** Late-k points rest on few surviving instances — the right tail is high-variance and should be read cautiously.

**why this form** A line over k is the only form that reveals the shape (rising vs flat) the profiles depend on.

### 7.5.4 CoT alignment

#### CoT alignment

`cot_alignment_by_model_domain`

**chart** grouped bars

**scope** Mean CoTaL (reasoning-vs-plan alignment) per model×domain.

**how to read** Higher = the reasoning trace structurally matches the emitted plan.

**data problems** Mostly NaN: most instances have no separate reasoning plan, so bars are absent or rest on tiny samples.

**why this form** Bars are fine where data exists; the dominant story here is missingness, documented in the CoT view.

#### Success rate by CoT flag

`cot_success_rate`

**chart** paired bars

**scope** Success rate split by whether CoT was present/enabled.

**how to read** A lift with CoT suggests reasoning helps; no lift suggests post-hoc rationalisation.  
**data problems** Confounded if CoT presence correlates with model family rather than being randomised.  
**why this form** A two-bar split is the minimal correct contrast for a binary flag.

#### CoT alignment: valid vs invalid

`cot_alignment_validity`

**chart** box / violin

**scope** Distribution of CoTaL for valid versus invalid plans.

**how to read** Higher alignment on valid plans = faithful reasoning; overlap = alignment doesn't track correctness.

**data problems** Needs enough valid AND invalid reasoning plans; the bundled run has few of either (18 both-valid, 8 both-invalid).

**why this form** A distribution contrast is needed because the claim is about separation, not means.

### 7.5.5 Composite & cross-domain

#### Success-rate heatmap

`success_rate_heatmap`

**chart** heatmap

**scope** Raw SR per model×domain — no normalisation, so cells are comparable across the whole grid.

**how to read** Empty columns = domains nobody solved (difficulty or infrastructure); empty rows = models that solved nothing.

**data problems** Genuinely sparse: many cells are 0, which is informative rather than missing.

**why this form** Unlike the rank heatmap, raw SR is on one common scale, so a single-scale heatmap is valid.

#### Failure-mode taxonomy

`failure_mode_taxonomy`

**chart** quadrant bubble

**scope** Halluc (x) vs PAS (y), bubble size = FASR, colour = SR — four named failure regimes.

**how to read** Locate each model in a quadrant; same-SR models in different quadrants fail differently. (See Cross-Domain view.)

**data problems** Bubbles collapse near an axis when a metric is 0/NaN for many models; small samples make positions unstable.

**why this form** Two orthogonal failure axes plus two encodings packs the entire failure diagnosis into one figure.

#### Composite Planning Score

`composite_scores`

**chart** bars + CI

**scope** PS per model with the 95% bootstrap CI.

**how to read** Sort by height for the leaderboard; overlapping CIs = the difference is not resolvable.

**data problems** Overall PS can sit outside its own per-row bootstrap CI for NaN-heavy models (kimi caveat).

**why this form** Bars with error bars are the standard, honest way to show a ranked scalar with uncertainty.

#### Within-domain rank heatmap

`domain_ranking_heatmap`

**chart** heatmap (faceted)

**scope** Per-domain normalised rank (0=best,1=worst) for every RANK\_METRIC.

**how to read** Compare cells DOWN a column WITHIN one domain block only — never across domain blocks.

**data problems** Normalisation is per-domain, so identical colours in different blocks are not comparable scores.

**why this form** Per-domain normalisation onto one colour scale is what lets many domains share a figure.

#### Rank variance

`rank_variance`

**chart** bars

**scope** Mean std of within-domain rank across domains, per model.

**how to read** Higher = domain specialist ( $\phi_4 \approx 2.39$ ); lower = generalist ( $\text{gpt\_oss} \approx 1.35$ ).

**data problems** Computed on raw ranks (not normalised) — magnitudes are run-specific; ordering is robust.

**why this form** A ranked bar is the right form for a single per-model dispersion scalar.

#### Domain correlation (Spearman)

`domain_correlation`

**chart** heatmap

**scope** Model×model Spearman  $\rho$  of per-domain success-rate vectors.

**how to read**  $\rho \approx 1$  = redundant probes; low/negative  $\rho$  = complementary models worth keeping both.

**data problems** Needs several domains with variance; with few domains and many SR=0 ties  $\rho$  is dominated by ties.

**why this form** A correlation matrix is the canonical view for pairwise similarity across a model set.

#### PS stacked by domain

`ps_by_domain_stacked`

**chart** stacked bars

**scope** Each model's PS split into per-domain contributions.

**how to read** Same total height can be one tall segment (concentrated) or many even ones (broad) — read the segmentation.

**data problems** Domains with no PS contribution simply don't appear; ensure the domain set is comparable across models.

**why this form** Stacking is the natural encoding for 'is this total broad or concentrated' — a pie would hide the total.

#### Metrics summary table

`metrics_summary_table`

**chart** table figure

**scope** All key aggregate metrics rendered as a matplotlib table for archiving.

**how to read** A static snapshot of the overall table; read it as the numeric source of the bar/heatmap figures.

**data problems** None beyond whatever NaNs the underlying tables carry; it faithfully shows them.

**why this form** A rendered table travels with the plot bundle so a figure folder is self-documenting.

#### Capability radar

`radar_chart`

**chart** radar

**scope** Polar pentagon of FASR · IWSR · Exec · IHR · PAS per model (base profile, no CoT bonus).

**how to read** Area = overall strength; shape = capability signature. A collapsed pentagon = No Grounding.

**data problems** Equal areas can hide very different shapes; many overlaid models become unreadable — facet for publication.

**why this form** A radar is justified for a small fixed set of bounded axes where the shape itself is the message.

#### Why the PNGs are not embedded here

The 27 figures are run-specific — they depend on which models and domains were in the bundle and would be stale the moment the suite is re-run. This catalogue documents what each figure *means and how to read it*



so it stays valid across runs; regenerate the actual PNGs with the pipeline against your own `results/plots/` directory.

## 8 Results and Conclusions

To conclude, it is worth highlighting the successful creation of a scalable and modular benchmark for evaluating the PDDL planning capabilities of Large Language Models. The framework was designed to be extensible to a wide range of problems and models, with a strong emphasis on producing easily interpretable and traceable results. The *raw*, *parsed*, *scored* output structure ensures diagnostic precision, allowing for a clear separation between generation, parsing, and validation failures.

### 8.1 General Conclusions on LLM Planning Capabilities

Our evaluation of several models highlighted critical insights about the current state of LLM-based planning: the main finding is that while LLMs often handle PDDL syntax and action vocabulary, their ability to perform robust, long-horizon, and goal-directed reasoning is still fragile. Models frequently generate action sequences that are syntactically correct and locally executable, as pointed out by generally high **Executability Ratios** and low **Hallucination Rates**, but these plans often fail to achieve the global goal, leading to low overall **Success Rates**. The significant gap between first-attempt success (**FASR**) and the final success rate (**SR**) highlights a strong dependency on iterative repair; this suggests that current models often behave more like retry-dependent generators than deterministic planners.

### 8.2 Limitations of the Study

It is also worth noting the limitations of this study: the evaluation is constrained by the available hardware, since these LLM models require significant resources to run effectively. The observed performance is also tied to the specific prompts and iterative repair strategies employed; different approaches could yield different results.

### 8.3 Hints for Future Work

The modularity of the benchmark creates several possibilities for future work and improvement:

- **Single Domain Testing:** The set of PDDL problems could be reduced to a single reasoning type and the difficulty increased gradually to see where a given LLM starts to fail at planning.
- **Advanced Interaction Protocols:** Future work could explore more sophisticated interaction protocols. This includes implementing more informative feedback mechanisms for the iterative repair loop, providing models with specific details about precondition failures (e.g., "action '(A)' failed because precondition '(P)' was false") to better assess their self-correction abilities.
- **Broader Model Evaluation:** The benchmark should be used to evaluate a broader and more diverse set of models, to build a more comprehensive landscape of LLM planning skills.
- **Single Model Release Evaluation:** The benchmark could be used to evaluate different releases of the same model, to test how its planning capabilities evolve across versions.
- **Deeper Analytical Insights:** The analytical layer can be extended to perform deeper correlational studies, such as linking specific model capability profiles to performance on domains with certain structural properties (e.g., high branching factor or sparse rewards), which could reveal fundamental strengths and weaknesses of different architectures.

## A Extended Evaluation Details

The main body keeps the evaluation section compact. This appendix preserves the implementation-level details, extended literature cards and full metric-family notes used during development of the benchmark.

### A.1 Extended Pipeline and Data Flow

|                 |                     |                  |                  |                     |
|-----------------|---------------------|------------------|------------------|---------------------|
| <b>3</b>        | <b>1</b>            | <b>8</b>         | <b>27</b>        | <b>7</b>            |
| artifact layers | canonical DataFrame | aggregate tables | registered plots | capability profiles |

#### A.1.1 The pipeline at a glance

Everything funnels into one DataFrame, `df_metrics`, the single source of truth for every `groupby` and every plot. Solid (teal) arrows are the data path; dashed (grey) arrows inject the PDDL ground truth needed to score plans (Figure 3).

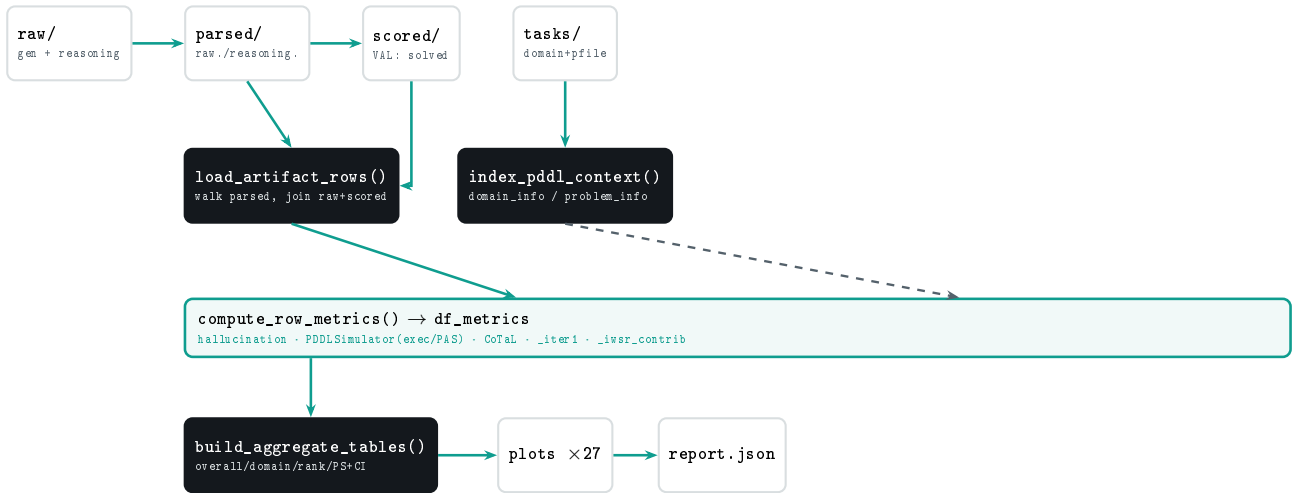


Figure 3: The analysis pipeline. One DataFrame, `df_metrics`, is the single source of truth; PDDL ground truth (dashed) is injected to score plans.

#### A.1.2 How a row is built: the three inputs to `df_metrics`

The canonical table is produced by `compute_row_metrics()`. It does *not* read the filesystem and it does *not* compute ground truth — both happen before it. By the time the row loop runs, three things already exist, each with a structurally different role.

| # | Input                                                | Produced by                                  | Role in the row loop                                                                                                                                                                                             |
|---|------------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <code>df_raw</code>                                  | <code>load_artifact_rows()</code>            | The skeleton: one row per instance with the artifact-derived columns and the keys used to look up ground truth. It is <code>.copy()</code> -ed, then <i>widened</i> with computed columns.                       |
| 2 | <code>domain_info</code> / <code>problem_info</code> | <code>index_pddl_context()</code>            | The ground truth: two dict lookups ( <code>d_info</code> , <code>p_info</code> ) consumed per row to score the plan. <i>Not</i> stored as columns — they are the reference the metric functions compare against. |
| 3 | <code>warnings_out</code>                            | threaded list via <code>add_warning()</code> | The diagnostic side-channel: a by-reference list accumulating structured records whenever an input is missing or malformed. Never part of <code>df_metrics</code> ; embedded separately in the report.           |

#### What `df_metrics` literally is

`df_metrics = df_raw.copy()` horizontally concatenated with three blocks of computed columns — hallucination, precondition/executability (from `PDDL Simulator`), and CoTaL — plus two derived columns `_iter1` and `_iwsr_contrib`. So input 1 supplies the columns you keep; input 2 is *consumed* to manufacture the new columns and then discarded; input 3 is a parallel log, not a column. The “three components” mental model is right, but only the first is literally part of the frame.

`df_raw` — the artifact skeleton `load_artifact_rows()` walks the canonical path `parsed/<run>/<model>/<protocol>/<domain>/<difficulty>/<instance>.json` and, for each instance, joins the matching `scored/` data. It emits exactly *one row per problem instance*, collapsing the retry loop by keeping the last non-empty plan as the representative `_actions` while preserving the full attempt lists as private columns.

```
# public -- identity + outcome
Model, Domain, Problem, Difficulty, Protocol, Run_id
Valid (= scored["solved"]), Length, Iterations (= scored["iterations_used"]), Chain_of_Thought
# private -- consumed downstream, dropped from the report
_actions          # representative plan (last non-empty attempt)
_cot_text          # reasoning trace text
_parsed_attempts _scored_attempts _raw_attempts  # full per-iteration lists
_file_path parsed_file_path
```

The keys that matter for the next input are `Domain`, `Difficulty` and `Problem` — together the exact triple used to retrieve per-instance ground truth. The `Chain_of_Thought` flag is resolved from three sources (protocol YAML, presence of reasoning text, presence of reasoning actions) so CoT-bearing runs are never mislabelled.

#### Why one row per instance, not per attempt

Keeping the grain at the *instance* level means every aggregate (success rate, hallucination, PS) is automatically an average over comparable units. The iteration structure is not lost — it survives in the private `*_attempts` columns and the `Iterations` count — but it is opt-in, read only by metrics that need it (FASR, IWSR, the iteration profile, CoTaL).

**The PDDL parsing layer** — `d_info` and `p_info` Plans can only be scored against the formal task they were generated for, so the framework parses the PDDL itself. Two small, deliberately permissive regex parsers do this — a syntax error emits a warning, never aborts the run. They answer two different questions and return two differently-shaped dicts.

`parse_domain_pddl(domain_path)` reads one `domain.pddl` (the schema shared by every instance in a domain) and returns the legal vocabulary plus per-action templates:

```
parse_domain_pddl(...) -> {
  "action_names": set[str],      # every legal action head -- the hallucination whitelist
  "predicates":   set[str],      # legal predicate names
  "functions":    set[str],      # numeric fluent names -- what is numeric, not propositional
  "schemas": {                  # one entry PER action, the simulator's instruction sheet
    action_name: {
      "params": [str, ...],      # ordered ?param variable names, for grounding
      "prec_raw": str,           # raw :precondition s-expression text (unparsed)
      "eff_raw": str,            # raw :effect s-expression text (unparsed)
    }, ...
  },
  "path": str | None,           # None when the file was missing (empty stub)
}
```

Each field has a single downstream consumer: `action_names` is the whitelist for action hallucination; `functions` tells the simulator which heads are numeric fluents; and `schemas` is what the simulator grounds — `params` maps the action's arguments onto the variables inside the raw `prec_raw` / `eff_raw` strings so preconditions can be checked and effects applied.

`parse_problem_pddl(problem_path)` reads one instance file (its own object set and starting state) and returns:

```
parse_problem_pddl(...) -> {
  "objects":      set[str],      # legal argument tokens -- object-hallucination whitelist
  "init_atoms":   set[tuple[str, ...]], # initial true propositions, e.g. ("at","truck1","depot")
  "init_numeric": dict[tuple, float],   # initial fluent values, e.g. ("fuel","truck1") -> 50.0
  "path":         str,
  "difficulty":   str,           # parent folder name
}
```

Object hallucination uses `objects`; the simulator seeds its world from `init_atoms` and `init_numeric`. Action- vs object-hallucination are kept as separate signals on purpose — a model can know the legal action names yet still invent objects, a different and shallower grounding failure.

`index_pddl_context(tasks_dir)` calls those two parsers exactly once each per file, walking `tasks/` a single time, and packs the results into two lookups whose key shapes mirror the granularity of what they hold:

```
domain_info : dict[str, dict]          # key = domain_name (schema is shared)
problem_info : dict[tuple[str,str,str], dict] # key = (domain, difficulty, instance)
```

This is the “parse once, consume many” design: every PDDL file is read once regardless of how many rows or retries reference it. In the row loop the retrieval is then two  $O(1)$  dict gets:

```
d_info = domain_info.get(domain, {})          # by domain
p_info = problem_info.get((domain, difficulty, instance), {}) # by full triple
pddl_ok = bool(d_info.get("action_names")) and bool(p_info.get("objects"))
```

#### A missing domain and a missing problem fail differently

When `domain.pddl` is absent, `index_pddl_context` still stores a populated-but-empty stub under the domain key (empty `action_names`, empty `schemas`, `path=None`). When a problem file is absent, no entry is created and the row’s `.get(..., {})` returns a bare `{}`. Both end up empty, but only the domain case leaves a record — and an empty `action_names` set means *every* emitted action looks illegal, silently depressing that domain’s grounding numbers. The only trace is a warning, which is exactly what input 3 is for.

`warnings_out` — the diagnostic accumulator `warnings_out` is a single `list[dict]` created once at the top of the analysis run and threaded by reference into every stage that can encounter a malformed or missing input — `index_pddl_context`, both PDDL parsers, and `compute_row_metrics`. Each stage appends through one helper:

```
def add_warning(warnings_out, warning_type, message, **extra):
    warnings_out.append({"type": warning_type, "message": message, **extra})
    # **extra carries locating context: run_id, domain, difficulty, problem, path
```

Its purpose is to replace scattered `print` / `logging` calls with one structured, serialisable trace that travels with the results: the full list is embedded in `report.json` under the `warnings` key, so a notebook or CI job can filter by `type` and locate the offending artifact without re-running anything. Typical entries are `missing_tasks_dir`, `missing_domain_pddl`, `missing_pddl_context`, `domain_parser_error`, `problem_parser_error`. Rows with incomplete context are *not* dropped — they still get NaN metric values — so the instance count stays honest while the warning records why those NaNs exist.

#### One diagnostic path bypasses `warnings_out`

The simulator is the exception. If `compute_precondition_metrics` throws, it falls back to a NaN result and reports through Python’s `stdlib` `warnings.warn("[simulator] ...")` — *not* through `add_warning`. So a simulator crash does not appear in the structured `warnings` list in the JSON report; it surfaces only on `stderr`. Worth unifying if the report is to be the single audit log.

### A.1.3 The merge: three inputs into one widened frame

`compute_row_metrics` iterates `df_raw` once. For each row it resolves `d_info` / `p_info` (input 2), runs the three metric families against the row’s `_actions` and attempt lists, and emits any incompleteness into `warnings_out` (input 3). The per-family results are collected and concatenated columnwise back onto the copied skeleton.

```
df = df_raw.copy()
for _, row in df.iterrows():
    d_info = domain_info.get(row["Domain"], {})
    p_info = problem_info.get((row["Domain"], row["Difficulty"], row["Problem"]), {})
    if not (d_info.get("action_names") and p_info.get("objects")):
        add_warning(warnings_out, "missing_pddl_context", ...) # input 3
    halluc_rows.append(compute_hallucination_metrics(actions, d_info, p_info))
    precondition_rows.append(compute_precondition_metrics(actions, d_info, p_info)) # PDDLsimulator
    cot_rows.append(... compute_cot_alignment_for_attempt(...) ...)
```

```
df_metrics = pd.concat([df, halluc_df, precondition_df, cot_df], axis=1)
df_metrics["_iter1"] = (df.Valid == True) & (df.Iterations == 1) # -> FASR
df_metrics["_iwsr_contrib"] = 1/Iterations if Valid else 0 # -> IWSR
```

The result, `df_metrics`, is the single source of truth every `groupby` in `build_aggregate_tables` and every plot reads from. Note the grain: the hallucination and executability columns describe the *representative* (last non-empty) plan, while the CoTaL and efficiency signals reach back into the preserved attempt lists.

#### A.1.4 Data classes that make the metrics possible

Two classes carry state that a flat function could not — introduced here, with the simulator expanded in full in Section A.1.5.

**PDDL Simulator (in the analysis script).** A forward STRIPS + simple-numeric simulator written in pure Python so the analysis layer needs no VAL subprocess. Instantiated *once per (problem, plan)* inside `compute_precondition_metrics`. Its decisive feature is `prop_hist`: a list of `frozenset` snapshots of the propositional state after every applied step. That history is what lets the framework, at the first failing action, ask “*was this precondition ever true earlier?*” — the question separating a sequencing error (true before, deleted too early) from a state fabrication (never true). Without the per-step history there is no PAS and no temporal distance.

**RunMetrics** (`evaluators/metrics.py`, `@dataclass`). The validator-side metric bundle written at scoring time: `validity_at_1`, `validity_at_k`, `repair_success`, `iterations_to_valid`, `plan_length`, `error_type`, `hit_iteration_limit`. The analysis script re-derives most of these from the scored JSON, but this dataclass is the contract guaranteeing every model is scored on the same basis, independent of backend.

The error vocabulary is fixed in `evaluators/error_taxonomy.py` as an `ErrorType` enum, partitioned into `LOGICAL_ERROR_TYPES` (empty plan, syntax, unknown action, invalid precondition, unsatisfied goal) versus `TECHNICAL_ERROR_TYPES` (parse error, timeout, validator crash/unavailable). That split keeps infrastructure failures from being charged against a model’s planning score.

#### A.1.5 PDDL Simulator in depth

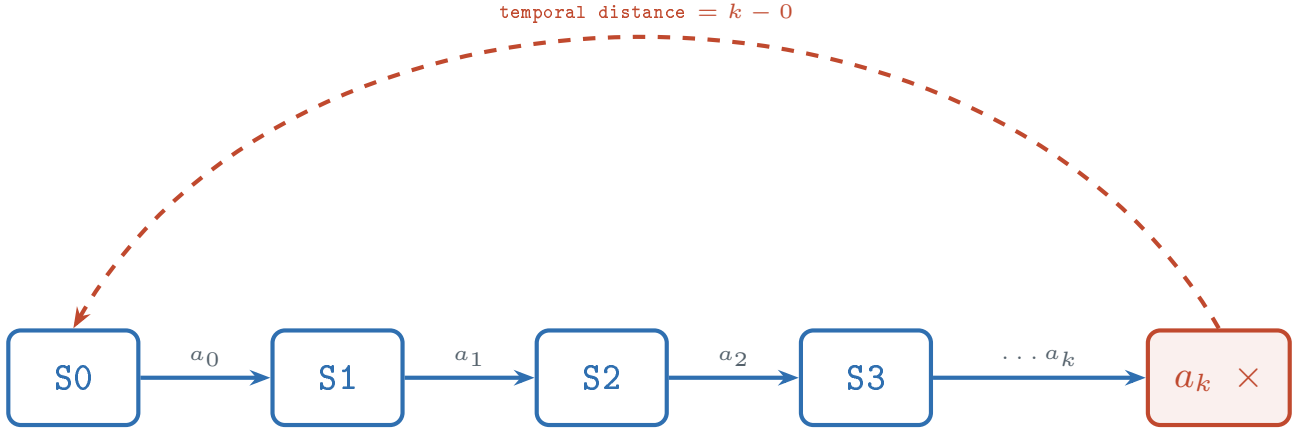
Almost every execution-side discriminator — executability ratio, the sequencing-vs-fabrication split, PAS, and mean temporal distance — is produced by one small class. It is a **forward STRIPS + simple-numeric simulator written in pure Python**, deliberately so: the analysis layer can re-derive these signals without spawning a VAL subprocess or shipping a compiled binary. It reproduces just enough of the validator’s logic for the propositional and numeric-inequality fragments the benchmark uses, trading completeness for portability.

**What it is for.** The class does three things: it **holds a world state** (a set of true propositions plus a dict of numeric fluents), it **walks the plan one action at a time** checking each action’s preconditions before applying its effects, and — its decisive feature — it **saves a frozen snapshot of the state after every applied step**. That snapshot history is what makes the precondition-awareness metrics computable. It is the basis for four metrics: executability ratio (how far the plan runs before the first break), the sequencing-error / state-fabrication counts, PAS (their ratio), and mean temporal distance. It targets *forward, propositional* and *simple-numeric* domains — STRIPS add/delete effects plus `increase` / `decrease` fluents and  $\geq \leq > <$  comparisons — and is invoked exactly once per `(problem, plan)` pair.

**State, snapshots, and the step loop.** The constructor seeds the world from the parsed problem and takes the first snapshot before any action runs; `prop_hist` then grows by one frozen set per applied step.

```
def __init__(self, d_info, p_info):
    self.prop = set(p_info.get("init_atoms", set())) # true propositions
    self.num = dict(p_info.get("init_numeric", {})) # numeric fluents
    self.prop_hist = [frozenset(self.prop)] # snapshot S0 = initial state
    self.step = 0
```

For each action, `compute_precondition_metrics` grounds the schema, checks preconditions, and — only if they hold — applies the effects and appends a new snapshot. The walk **stops at the first failing action** (a `break`), which makes “how far did it get” a meaningful prefix length.



`prop_hist = [frozenset(S0), frozenset(S1), ...]` — one snapshot per applied step

Figure 4: State evolution. At the first failing action  $a_k$ , each unmet precondition atom is traced back through `prop_hist`: found earlier  $\rightarrow$  **sequencing error** (record  $k - \text{last}$  as temporal distance); never found  $\rightarrow$  **state fabrication**.

```
for step, action_str in enumerate(actions):
    name, args = parse_action(action_str)
    schema = schemas.get(name)
    if name is None or schema is None:      # unparseable / unknown action: skipped
        continue
    if len(schema["params"]) != len(args):  # wrong arity -> fabrication, stop
        first_failure = step; fab_errors += 1; break
    ok, failed_prop, failed_num = sim.check_preconditions(schema["prec_raw"], params, args)
    if not ok:
        first_failure = step
        for atom in failed_prop:
            last = sim.last_step_atom_true(atom)      # search the snapshot history
            if last >= 0: seq_errors += 1; temporal_distances.append(step - last)
            else: fab_errors += 1
        fab_errors += len(failed_num); break
    sim.apply_effects(schema["eff_raw"], params, args) # mutate state + append snapshot
```

**The classification logic, atom by atom.** At the failing action, each unsatisfied propositional precondition is the unit of analysis. `last_step_atom_true(atom)` scans `prop_hist` newest-to-oldest and returns the latest snapshot index at which the atom held — or `-1` if it never did.

```
def last_step_atom_true(self, atom):
    for index in range(len(self.prop_hist) - 1, -1, -1):
        if atom in self.prop_hist[index]:
            return index
    return -1
```

That single lookup is the whole basis of the sequencing/fabrication distinction. If the atom was true at some earlier snapshot, the model had the prerequisite and then ordered actions so as to destroy it — a **sequencing error**, and the gap  $\text{step} - \text{last}$  is recorded as a temporal distance. If the atom was never true, the model required a world state that never existed — a **state fabrication**. Effects are applied in STRIPS order (deletes before adds), so the snapshots faithfully reflect what each action left behind. The metrics fall straight out of these counts:

```
executability_ratio = first_failure / plan_len
PAS = sequencing_errors / (sequencing_errors + state_fabrications)
mean_temporal_distance = mean(step - last_true_step for each sequencing error)
```

#### The counts describe the first failure, not the whole plan

Because the loop `break`s at the first failing action, the sequencing/fabrication counts and the temporal distances come from that *one* action's unmet preconditions — PAS is a property of where the plan first breaks, not a tally over the entire sequence. This is intended (it localises the failure), but it means a plan can accrue at most one failure event in these columns.

#### Unknown actions are skipped, which inflates executability

An action whose name is not in the schema is `continue`d over — it neither advances the state nor counts as a failure here (name-level hallucination is scored separately). So a plan made entirely of hallucinated action names never triggers a precondition failure and records `executability_ratio = 1.0`. Always read ER alongside the hallucination columns. Relatedly, because skipped actions advance the plan index but not the snapshot index, temporal distance is measured in generated-plan positions relative to the snapshot where the atom last held — the two clocks coincide only when every prior action was a known, applied schema.

### A.1.6 From data to metrics to pictures — why each stage exists

The architecture separates concerns on purpose, so each is independently testable.

| Stage       | Function                            | Purpose / why it is separate                                                                                                     |
|-------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Load        | <code>load_artifact_rows</code>     | Decouples the filesystem walk from metric logic; everything below receives a uniform table.                                      |
| Row metrics | <code>compute_row_metrics</code>    | One pass attaches every per-instance metric; private <code>_actions</code> / <code>_cot_text</code> stay alive only here.        |
| Aggregate   | <code>build_aggregate_tables</code> | All <code>groupby</code> logic in one place so the JSON report can embed tables whether or not plots are drawn.                  |
| Visualise   | <code>maybe_make_plots</code>       | Reads only the aggregate tables; registering each figure embeds its path + description into the report.                          |
| Assemble    | <code>build_model_payloads</code>   | Re-keys everything by <i>model</i> so a consumer can pull one model in O(1), and attaches the capability-profile classification. |

#### A real artefact to know about

Because `compute_row_metrics` keeps the last non-empty plan as the representative for the instance-level hallucination/executability columns, those columns describe the *final* attempt. Metrics that must see all attempts (FASR, IWSR, CoTaL per-iteration, the iteration profile) read the preserved `_*_attempts` lists instead. When reading a plot, always check which grain it is on — this is spelled out per metric in the following sections.

## A.2 Extended Scientific Foundation

The ten papers that license the metric set — each pinned to the exact paragraph, table, figure or definition it is taken from, with numbers, paraphrased, linked, and mapped to the design decision it grounds; plus the literature-motivated metrics not yet implemented.

The framework’s thesis is that binary validity is not enough: two models can both fail and be failing for completely different reasons. Every metric isolates one mechanism, and each is anchored to a specific result in the literature. Each card below pins the *exact* location the claim is taken from, paraphrases what that spot reports (with its numbers), states what it grounds here, and links to the source.

### The one-line argument

If LLM planning is closer to approximate retrieval than to search (Kambhampati 2024; Valmeekam 2023), then (a) retries are near-independent samples so success must be discounted by retry count —

*FASR, IWSR*

; (b) feedback content is largely irrelevant so a large retry gap is not correction —

*Stechly 2023*

; and (c) the only way to separate retrieval from reasoning is to probe execution, grounding and reasoning-faithfulness directly —

*exec\_ratio, PAS, hallucination, CoTaL*

.

### Provenance and a corrected citation

Locations, page numbers and quantitative results below follow the project’s verified reference apparatus. The Critical Investigation is cited from the NeurIPS 2023 proceedings (the version of record) with preprint arXiv:2302.06706 — this supersedes the arXiv:2305.15771 id used in an earlier pass. The 2022 benchmark ("Still Can't Plan", arXiv:2206.10498) and the 2025 frontier study (Correa et al., arXiv:2511.09378) are added here. Findings are paraphrased; quoted fragments are short technical phrases; numbers are reported as facts.

### REFERENCE: Large Language Models Still Can't Plan (A Benchmark for LLMs on Planning and Reasoning about Change)

Valmeekam, Olmo, Sreedharan & Kambhampati (2022) · *NeurIPS 2022 FMDM Workshop* arXiv:2206.10498  
↗ DOI 10.48550/arXiv.2206.10498

**Where in the paper** §5 "Current Curriculum for Testing", the seven test cases (pp. 6–8); §5.2 Optimal Planning + Table 1; §7 Conclusion (p. 9).

**What that location reports** Introduces a seven-task curriculum — plan generation, cost-optimal planning, reasoning about plan execution, robustness to goal reformulation, plan reuse, replanning, plan generalisation — precisely because plan-validity alone captures only one dimension; only the first two are "actual planning", the rest are auxiliary reasoning-about-change tasks. InstructGPT-3 solved ~ 3.2% of optimal-planning vs ~ 5% of plan-generation instances (valid ≠ efficient). The conclusion explicitly lists adding a partial-correctness metric as future work.

**What it grounds here** The original literature justification for going beyond pass/fail — the framework’s whole thesis. Motivates the (not-yet-implemented) partial-credit / Goal-Achievement score and an Optimality Ratio.

**Read it** Open the source ↗

### REFERENCE: On the Planning Abilities of Large Language Models — A Critical Investigation

Valmeekam, Marquez, Sreedharan & Kambhampati (2023) · *NeurIPS 2023* NeurIPS 2023 · arXiv:2302.06706  
↗ DOI 10.48550/arXiv.2302.06706

**Where in the paper** §2 Related Work para.5 (p. 3) Phase 1/Phase 2; §4 Autonomous Mode" Obfuscating names..."(pp. 7–8) + Table 1; §4 Figure 3 (p. 7) relaxations; §5.2 "Verifier-assisted repeated backprompting", Table 4 (p. 9); Table 1 CoT rows (p. 5).

**What that location reports** Splits planning into Phase 1 (discovering actions and their precondition/effect dependencies) and Phase 2 (sequencing them without harmful interactions); LLMs are strong on Phase 1, weak on Phase 2. Renaming Blocksworld into Mystery Blocksworld collapses GPT-4 zero-shot from 210/600 (35%) to 1/600 (0.16%) — "pattern matching rather than reasoning". Figure 3 classifies failures as inexecutable vs executable-but-goal-not-reached. Backprompting (≤15 rounds) eventually solves 82% Blocksworld / 70% Logistics at avg 3.68 / 3.31 rounds, but only 10% in Mystery Blocksworld. CoT barely helps (Blocksworld 210→214/600; Mystery 1→54/600).



**What it grounds here** Grounds the `exec_ratio` + PAS pair (Figure 3's classification, refined into sequencing vs fabrication), the hallucination gate (obfuscation collapse), FASR (pre-backprompt performance), cross-domain consistency (the obfuscation gap), the Lucky-Retriever profile, and CoT as diagnostic not performative.

**Read it** [Open the source ↗](#)

#### REFERENCE: GPT-4 Doesn't Know It's Wrong: An Analysis of Iterative Prompting for Reasoning Problems

*Stechly, Marquez & Kambhampati (2023) · NeurIPS 2023 FMDM Workshop* [arXiv:2310.12397 ↗](#) DOI 10.48550/arXiv.2310.12397

**Where in the paper** Abstract, findings (i)–(iii); Abstract para.2 (the complexity-argument paragraph).

**What that location reports** On graph colouring GPT-4 is bad at solving and no better at verifying, so LLM-critiques-LLM loops do not help; crucially the content of the criticisms — from an LLM or an external solver — is "largely irrelevant" to iterative-prompting performance. The intuition that verification is easier than generation is a complexity argument that does not apply to approximate retrieval. Apparent iterative gains come from the correct answer fortuitously appearing across repeated independent samples.

**What it grounds here** The keystone of the retry apparatus: if feedback is irrelevant, each retry is an independent re-sample → a large Retry Gap is stochastic search, IWSR discounts late wins, and a flat  $P(\text{Valid}|\text{k})$  is the Stochastic-Searcher signature.

**Read it** [Open the source ↗](#)

#### REFERENCE: Can Large Language Models Reason and Plan?

*Kambhampati (2024) · Annals of the New York Academy of Sciences 1534(1):15–18* [arXiv:2403.04121 ↗](#) DOI 10.1111/nyas.15125

**Where in the paper** para.4–7 the two-phase model; pp. 3–4 the self-critique paragraphs; p. 2 the approximate-retrieval / CoT paragraph.

**What that location reports** Distinguishes having planning-domain knowledge (Phase 1) from assembling it into an executable plan that handles subgoal interactions (Phase 2). What LLMs do is "universal approximate retrieval", not principled reasoning; self-critique is of questionable utility because they hallucinate both false positives and false negatives when verifying their own solutions. Hence the fake-reasoning vs genuine-deliberation distinction.

**What it grounds here** The theoretical organiser of the failure-mode quadrant; the justification for IWSR penalising late convergence; and the SR-high + CoT-low "fake reasoning" diagnosis.

**Read it** [Open the source ↗](#)

#### REFERENCE: Language Models as Zero-Shot Planners: Extracting Actionable Knowledge for Embodied Agents

*Huang, Abbeel, Pathak & Mordatch (2022) · ICML 2022 · PMLR 162:9118–9147* [arXiv:2201.07207 ↗](#) DOI 10.48550/arXiv.2201.07207

**Where in the paper** §2.2 "Metrics" and Figure 1 (pp. 2–3).

**What that location reports** First formal separation of executability from correctness as two independent axes: large models (GPT-3 175B, Codex 12B) produce plans with high semantic correctness but low executability — as low as ~ 18% for raw Codex. A wander-around program is highly executable but incorrect; a human-written plan is correct but often inexecutable — so both axes are needed.

**What it grounds here** The direct basis for `exec_ratio` as a metric separate from `Success_Rate`; the framework's `exec_ratio` is a continuous, finer-grained version of Huang's binary executability.

**Read it** [Open the source ↗](#)

#### REFERENCE: Measuring Faithfulness in Chain-of-Thought Reasoning

*Lanham, Chen, Radhakrishnan et al. (2023) · Anthropic (2023)* [arXiv:2307.13702 ↗](#) DOI 10.48550/arXiv.2307.13702

**Where in the paper** Abstract + §3 (the Early-Answering intervention and post-hoc-reasoning definition).

**What that location reports** Truncating the chain of thought before the answer tests whether the stated reasoning is causal or post-hoc; models vary widely in how much they condition on the CoT, and faithfulness tends to decrease as models scale.

**What it grounds here** CoTaL operationalises faithfulness as content/structure overlap between the reasoning trace and the emitted plan; the SR-high + CoT-low = fake-reasoning diagnosis is the direct

application.

**Read it** [Open the source ↗](#)

**REFERENCE: Embers of Autoregression: Understanding LLMs Through the Problem They Are Trained to Solve**

*McCoy, Yao, Friedman, Hardy & Griffiths (2024) · PNAS (2024) arXiv:2309.13638 ↗ DOI 10.48550/arXiv.2309.13638*

**Where in the paper** Abstract (cipher example); §5 sensitivity to task probability; §6 sensitivity to output probability; Table 1.

**What that location reports** Even on deterministic tasks, accuracy tracks the probability of the task / output / input. GPT-4 decodes a shift cipher ~ 51% of the time when the answer is high-probability but only ~ 13% when low-probability.

**What it grounds here** Grounds the FASR-by-difficulty analysis: a sharp easy→hard cliff signals sensitivity to training-data frequency, not logical complexity.

**Read it** [Open the source ↗](#)

**REFERENCE: Faith and Fate: Limits of Transformers on Compositionality**

*Dziri, Lu, Sclar et al. (2023) · NeurIPS 2023 (Spotlight) arXiv:2305.18654 ↗ DOI 10.48550/arXiv.2305.18654*

**Where in the paper** The computation-graph formulation of compositional tasks and the error-by-depth analysis.

**What that location reports** Multi-step tasks (multi-digit multiplication, a logic puzzle, a DP problem) are cast as computation graphs; transformers reduce multi-step reasoning to "linearised subgraph matching", with success tied to having seen graph fragments in training and accuracy decaying as compositional depth grows (e.g. ~ 59% on 3×3 multiplication).

**What it grounds here** Grounds the temporal-distance reading and the exec-vs-length decay slope (local steps fine, long-horizon sequencing collapses) — and the Understander profile, where the collapse depth is itself measurable.

**Read it** [Open the source ↗](#)

**REFERENCE: Evaluating the World Model Implicit in a Generative Model**

*Vafa, Chen, Rambachan, Kleinberg & Mullainathan (2024) · NeurIPS 2024 (Spotlight) arXiv:2406.03689 ↗ DOI 10.48550/arXiv.2406.03689*

**Where in the paper** Abstract + §2: the DFA / Myhill-Nerode framework, the MNI and MNB constructions, and the two metrics.

**What that location reports** Treats a deterministic domain (PDDL planning is a DFA) and asks whether a model's implicit world model is coherent. Two states are equivalent iff all future sequences agree; the Myhill-Nerode interior (MNI) is the set of continuations valid from both states, the boundary (MNB) the minimal suffixes that separate them. From these: a compression score (do same-state histories yield an empty implied boundary?) and a distinction score (does the model reproduce the true boundary between different states?). Models look fine on standard diagnostics yet their world models are "far less coherent than they appear".

**What it grounds here** Grounds the Vocabulary-Only profile (high exec / low PAS = surface fluency over an incoherent world model) — and is the strong form of which PAS is explicitly the weak version.

**Read it** [Open the source ↗](#)

**REFERENCE: The 2025 Planning Performance of Frontier LLMs (rev.: Frontier LLMs Rival State-of-the-Art Planners)**

*Correa, Pereira et al. (2025) · preprint (2025) arXiv:2511.09378 ↗ DOI 10.48550/arXiv.2511.09378*

**Where in the paper** Abstract, results tables and Figure 1; eight domains of the IPC 2023 Learning Track; plans VAL-verified.

**What that location reports** The most directly comparable contemporary benchmark. Evaluates DeepSeek R1, Gemini 2.5 Pro and GPT-5 against the classical planner LAMA; on standard PDDL tasks GPT-5 is competitive with LAMA. Under obfuscation (predicate/action names stripped of semantics) every LLM degrades while LAMA is invariant to renaming (GPT-5 152, Gemini 2.5 Pro 146, DeepSeek R1 93 solved). The expanded version adds Gemini 3/3.1 Pro (245 vs 234 baseline) and notes some obfuscated re-identification, hinting performance is not wholly reducible to training exposure.

**What it grounds here** Its Figure 1 is essentially the target plot. Introduces the standard-obfuscated

gap as a metric (small gap = structural reasoning, large gap = semantic exploitation) — a cleaner operationalisation of the hallucination/grounding axis, and endorses VAL verification + freshly-generated tasks against contamination.

**Read it** [Open the source](#) ↗

### A.2.1 Claim → paper → location → support

The spine of the design: each implementation decision tied to the precise spot that licenses it, with the supporting result paraphrased.

| Framework claim                              | Paper            | Location                         | Support                                                              |
|----------------------------------------------|------------------|----------------------------------|----------------------------------------------------------------------|
| Binary validity is insufficient              | Valmeekam 2022   | §5 seven-task curriculum         | Only 2 of 7 tasks are "actual planning"                              |
| Partial credit / optimality needed           | Valmeekam 2022   | §7; §5.2 Table 1                 | Partial-correctness flagged as future work; 3.2% optimal vs 5% valid |
| Hallucination gate threshold                 | Valmeekam 2023   | §4 Obfuscation, Table 1          | GPT-4 35% → 0.16% under name obfuscation                             |
| Phase 1 / Phase 2 distinction                | Valmeekam 2023   | §2 Related Work, p. 3            | discovering actions vs sequencing a subset                           |
| exec_ratio + PAS pair                        | Valmeekam 2023   | §4 Figure 3                      | inexecutable vs goal-not-reached classification                      |
| FASR as honest metric                        | Valmeekam 2023   | §5.2 Table 4, p. 9               | 82%/70% eventual, avg 3.68/3.31 rounds; Mystery 10%                  |
| Lucky-Retriever profile                      | Valmeekam 2023   | §4 Obfuscation                   | "pattern matching rather than reasoning"                             |
| Stochastic Searcher / retry gap              | Stechly 2023     | Abstract, finding (iii)          | criticism content "largely irrelevant"                               |
| IWSR penalises late retries                  | Stechly 2023     | Abstract para.2                  | approximate retrieval voids the complexity argument                  |
| Fake reasoning (SR high + CoT low)           | Kambhampati 2024 | p. 2 approximate retrieval       | retrieval vs deliberation                                            |
| exec & SR independent axes                   | Huang 2022       | §2.2 Metrics, Figure 1           | executability/correctness trade-off                                  |
| CoT alignment metric                         | Lanham 2023      | Abstract + §3                    | CoT faithfulness operationalised                                     |
| Cliff drop easy → hard                       | McCoy 2024       | Abstract + §5–6                  | performance follows training probability                             |
| High exec + low PAS = incoherent world model | Vafa 2024        | Abstract + §2 (MNI/MNB)          | surface fluency without a coherent world model                       |
| Long-horizon collapse (Understander)         | Dziri 2023       | computation-graph error-by-depth | reasoning decays with compositional depth                            |
| Standard - obfuscated gap                    | Correa 2025      | Results, Figure 1                | LLMs degrade under renaming; LAMA invariant                          |

### A.2.2 Literature-motivated metrics beyond the current set

The same papers also point to metrics the framework does not yet implement. Listed here as honest future directions, not as shipped features.

#### On DOIs and versions of record

arXiv preprints resolve through 10.48550/arXiv.<id>; Kambhampati 2024 (Annals NYAS, 10.1111/nyas.15125) and McCoy 2024 (PNAS) also carry publisher DOIs. The Critical Investigation links to the NeurIPS proceedings PDF; Huang 2022 is canonically PMLR 162 (preprint arXiv:2201.07207). Correa et al. 2025 (arXiv:2511.09378) has a v1 title ("The 2025 Planning Performance...") and an expanded later version ("...Rival State-of-the-Art Planners"); both share the id.

| Candidate metric                        | Source              | What it would add                                                                                                      |
|-----------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------|
| Goal-Achievement / partial-credit score | Valmeekam 2022 §7   | Fraction of goal atoms achieved, instead of all-or-nothing validity — the benchmark authors' own stated gap.           |
| Optimality Ratio                        | Valmeekam 2022 §5.2 | Plan length / optimal length, so a long valid plan is not scored equal to a minimal one.                               |
| Standard - obfuscated gap               | Correa 2025         | Per-model delta between named and renamed domains — a direct, cleaner grounding measure than hallucination rate alone. |
| Strong world-model score (MNI/MNB)      | Vafa 2024           | Compression + distinction scores over shared mid-plan sub-states; PAS is the weak, single-step version of this.        |

## A.3 Extended Metric Families

### A.3.1 Planning Efficiency

Success Rate, First-Attempt Success Rate, Iteration-Weighted Success Rate, Retry Gap — and the iteration profile that distinguishes genuine correction from stochastic resampling.

This family answers "*how often, how cheaply, and how honestly does the model succeed?*" All four metrics live in the `overall` / `by_domain` tables and share one design principle: a success that needed many retries is worth less than a success on the first try, because under the retrieval hypothesis those retries are independent samples, not corrections.

#### Real values from the bundled run

gpt\_oss\_120b is the clear leader: SR  
0.524, FASR  
0.333, IWSR  
0.371, Retry  
Gap  
0.190. qwen\_3\_5\_122b shows the diagnostic pattern that motivates the whole family — SR  
0.500 but FASR  
0.000 and Retry  
Gap  
0.500: every success arrived only after retries, the signature of a Stochastic Searcher.

#### METRIC · SR: Success Rate

`overall["Success_Rate"] = mean(Valid)` Fraction of instances the model ever solves, across all allowed iterations. The headline number — and the one the framework argues is least trustworthy on its own.

**How it is computed** Boolean `Valid` (from `scored["solved"]`) averaged per group. This is *validity@k*: success anywhere in the repair budget counts.

```
Success_Rate=("Valid", "mean")
```

**Range & meaningful bands** Bands used by the profile thresholds:  $\approx 0 \rightarrow$  No Grounding ·  $< 0.2 \rightarrow$  Understander/Vocabulary ·  $> 0.5$  capable ·  $> 0.7$  strong. Always read against FASR; SR alone cannot distinguish retrieval from reasoning.

**In the plots** `success_rate_heatmap` (model×domain), the SR bars in `sr_vs_fasr` and `sr_fasr_iwsr_by_model`, and it drives the colour in `failure_mode_taxonomy` and `fasr_iwsr_scatter`.

**Rationale** SR is the necessary baseline but the framework's thesis is that it is *insufficient*: it conflates first-try success with retry-dependent success and says nothing about *why* a failure happened. Every other metric exists to decompose it.

#### METRIC · FASR: First-Attempt Success Rate

`overall["FASR"] = mean(_iter1)` Fraction solved on iteration 1 with no feedback at all. The framework treats this as the single honest signal of planning ability, because it is the only one that cannot be inflated by retries.

**How it is computed** A private boolean column is set at row level and averaged:

```
df["_iter1"] = (df["Valid"] == True) & (df["Iterations"] == 1)
FASR=("_iter1", "mean")
```

**Range & meaningful bands**  $FASR > 0.50$  with low Retry Gap  $\rightarrow$  genuine zero-shot capability.  $FASR \approx 0$  while SR is high  $\rightarrow$  the model only ever succeeds by resampling (the entire qwen / nemotron pattern in the bundled run).

**In the plots** `fasr_by_model_domain`, `fasr_by_difficulty`, the FASR bar in `sr_vs_fasr` / `sr_fasr_iwsr_by_model`, both axes of `fasr_iwsr_scatter`, and the y-axis of `failure_mode_taxonomy` bubble size.

**Rationale** Under the retrieval hypothesis (Kambhampati 2024) and the irrelevant-feedback finding (Stechly 2023), only the un-retried attempt reflects what the model can do in one shot. FASR carries the highest weight in the composite score (0.25) for exactly this reason.

#### METRIC · IWSR: Iteration-Weighted Success Rate

*contrib* =  $(1/\text{Iterations})$  if Valid else 0 Success discounted by how many attempts it took. Rewards models that converge fast; penalises models that only succeed deep into the retry budget.

**How it is computed** Each solved instance contributes  $1/k$  where  $k$  is the iteration count; unsolved contributes 0. The mean of these contributions is IWSR.

```
df["_iwsr_contrib"] = (1.0/row["Iterations"]) if row["Valid"] and row["Iterations"]>0 else 0.0
IWSR=("_iwsr_contrib", "mean")
```

**Range & meaningful bands** IWSR is bounded above by SR and below by FASR (a first-try win contributes the full 1.0, matching FASR; a 5th-try win contributes only 0.2).  $IWSR \approx SR \rightarrow$  fast repairs (Efficient Corrector).  $IWSR \ll SR \rightarrow$  slow, late convergence.

**In the plots** `iwsr_by_model_domain`, the IWSR bar in `sr_fasr_iwsr_by_model`, and the y-axis of `fasr_iwsr_scatter`.

**Rationale** The  $1/k$  weighting is the numerical encoding of Stechly's argument: if verification isn't easier than generation for a retrieval system, late successes deserve no credit for "search". IWSR is the metric that turns the retry budget into a cost.

#### METRIC · RG: Retry Gap

*Retry\_Gap* = *Success\_Rate* - *FASR* How much of the reported success depends on retries. A pure diagnostic, not a capability — high is a warning, not a strength.

**How it is computed** A single subtraction on the `overall` / `by_domain` tables, then sorted ascending in the `retry_gap` table.

```
overall["Retry_Gap"] = overall["Success_Rate"] - overall["FASR"]
```

**Range & meaningful bands**  $RG < 0.10 \rightarrow$  near-zero-shot.  $0.10\text{--}0.30 \rightarrow$  Efficient Corrector territory.  $RG > 0.30 \rightarrow$  Stochastic Searcher: success is mostly resampling. In the bundled run `qwen_3_5` hits 0.500.

**In the plots** `retry_gap_by_model` (diverging horizontal bars, red = positive gap). RG is also a ranked metric (direction = min) in the within-domain ranking.

**Rationale** RG operationalises the difference between "solved" and "solved on its own". It is the cleanest single read on whether a model's SR is trustworthy.

### A.3.2 The iteration profile — the test that separates correction from resampling

RG tells you retries happened; the iteration profile tells you *whether they helped*. `compute_iteration_profile()` computes, for each attempt slot  $k$ , the conditional success probability among instances that needed exactly  $k$  attempts:

```
for k in range(1, max_iter+1):
    reached = df[df["Iterations"] >= k]
    exact = reached[reached["Iterations"] == k]
    p_valid_given_reached = exact["Valid"].mean()
```

A **rising** curve means later attempts succeed more often — the model is using feedback (Efficient Corrector). A **flat** curve means success probability is independent of attempt number — each retry is an independent draw (Stochastic Searcher). `profile_is_flat()` formalises "flat" as  $\max\text{-min} \leq 0.10$  over the finite points, and feeds the Stochastic-Searcher profile condition. This is the direct empirical test of Stechly's finding (iii) in your own data.

#### Reading caveat

With few instances per (model,  $k$ ) slot the curve is noisy and `profile_is_flat`

can return None (inconclusive) when fewer than two finite points exist. Treat the profile as suggestive, not decisive, on small task sets — the bundled run's 6-instance subsets are exactly this regime.

### A.3.3 Reading the efficiency plots

The three success metrics are only diagnostic in combination. The guarantee  $\text{FASR} \leq \text{IWSR} \leq \text{SR}$  holds by construction, and the two gaps between them carry the meaning: the  $\text{SR}-\text{IWSR}$  gap measures how late successes arrive, the  $\text{IWSR}-\text{FASR}$  gap measures how much retries contributed at all.

**FASR by model  $\times$  domain.** Two readings live in one chart. **Per model (compare a model’s bars across domains):** a systematic gap between one domain group and another suggests memorised patterns specific to that group rather than generalised planning; a small gap across all domains points to genuine zero-shot transfer. A uniformly low FASR means near-zero zero-shot ability — go to SR-vs-FASR to check whether success is being manufactured by resampling. **Per domain (compare one domain across models):** a domain with consistently high FASR across every model is simply an easy domain — the domain, not the model, is driving the score.

| Gap pattern                                 | Meaning                                                                                                    |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------|
| $\text{SR} \approx \text{FASR}$ , both high | Genuine zero-shot capability. Retries add almost nothing.                                                  |
| $\text{SR}$ moderately $>$ $\text{FASR}$    | Partial capability, partly scaffolded by the retry mechanism.                                              |
| $\text{SR} \gg \text{FASR}$ (large gap)     | Stochastic search. Per Stechly et al. (2023), retries are re-sampling, not correction — SR alone misleads. |
| Both near 0                                 | No planning capability; the retry budget is irrelevant.                                                    |

### SR vs FASR — the retry gap.

**FASR by difficulty bin — genuine reasoning vs memorisation.** An easy $\rightarrow$ hard sweep separates reasoning from training-frequency exploitation. A reasoner degrades *gradually and smoothly* as steps, objects and constraint interactions grow — the decline is a property of the task. A memoriser stays high on easy (short, common-looking plans) and then falls off a *cliff* at the point where the required plans become long and rare, i.e. fall outside its training distribution. The cliff is the boundary of the distribution, not of a reasoning process.

| FASR pattern (easy $\rightarrow$ hard) | Interpretation                                                                                                          | Reference                      |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Gradual decline                        | Generalisation — graceful degradation under complexity; some genuine reasoning that scales.                             | Expected for capable reasoners |
| Sharp cliff to $\sim 0$                | Memorisation — easy instances exploit training frequency; hard instances fall outside the distribution.                 | McCoy et al. 2024              |
| Flat and high                          | Genuine capability across all difficulties. Strongest possible result.                                                  | Ideal case                     |
| Flat and low                           | No capability at any difficulty — or Difficulty=‘unknown’ throughout (check data).                                      | Valmeekam et al. 2023          |
| Non-monotonic / irregular              | Difficulty bins don’t track real plan complexity. Recalibrate using optimal-plan-length quantiles / statistical rarity. | Data-quality issue             |

**IWSR — where in the run the success happened.** The closer IWSR sits to FASR, the more successes landed on attempt 1; the closer it sits to SR, the faster the model converges when it does eventually succeed. The two gaps read together:

| SR-IWSR | IWSR-FASR | Interpretation                                                                                                                      | Profile                   |
|---------|-----------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| small   | small     | All three nearly equal; most success on attempt 1, retries add nothing.                                                             | Genuine zero-shot planner |
| small   | large     | $\text{IWSR} \approx \text{SR} \gg \text{FASR}$ — rarely first-try, but converges quickly on attempt 2-3.                           | Efficient corrector       |
| large   | small     | $\text{FASR} \approx \text{IWSR} \ll \text{SR}$ — most success arrives very late, at attempt 4-5, and is heavily penalised by IWSR. | Stochastic / late luck    |
| large   | large     | All three spread apart; SR is inflated by retries, while FASR reveals that true zero-shot performance is low.                       | Fully retry-dependent     |

**P(Valid | reached attempt k).** For each k the curve takes plans that actually *reached* attempt k ( `Iterations`  $\geq$  k ) and reports the fraction resolving at exactly k — a conditional, not cumulative, probability. Its shape is the direct empirical test of the Stochastic-Searcher hypothesis.

| Line shape         | Interpretation                                                                                                    | Reference                        |
|--------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Declining from k=1 | Easiest problems resolve first; harder ones take more attempts — difficulty-sensitive correction.                 | Expected for a capable reasoner  |
| Flat across k      | Each retry is statistically independent — re-sampling, not correcting. Feedback content is irrelevant.            | Stechly et al. 2023 (§4 Results) |
| Increasing with k  | Model improves with each retry — genuine in-context learning. Very rare; check whether feedback leaks the answer. | Would contradict Stechly et al.  |
| Low-flat near 0    | No valid plans at any attempt; the iteration budget is wasted compute.                                            | Valmeekam et al. 2023 (§5.2)     |

**Retry-gap bar & the FASR–IWSR scatter.** The diverging retry-gap bar colours each model by `SR - FASR` (tomato for the usual positive gap; a steelblue/negative bar would be a data anomaly, since  $SR \geq FASR$  by construction). Models at the top — lowest gap — are the most independent of the retry scaffold. The FASR-vs-IWSR scatter then encodes three axes at once:  $\mathbf{x} = \mathbf{FASR}$  (zero-shot skill),  $\mathbf{y} = \mathbf{IWSR}$  (retry efficiency),  $\mathbf{dot\ size + colour} = \mathbf{SR}$  (deep-red small = SR 0  $\rightarrow$  deep-green large = SR 1). No point can fall below the FASR=IWSR diagonal.

| Position                      | Dot                | Interpretation                                                                        |
|-------------------------------|--------------------|---------------------------------------------------------------------------------------|
| Top-right, on/near diagonal   | large green        | Genuine planner — high FASR, $IWSR \approx FASR$ , high SR; success is first-attempt. |
| Top-left, well above diagonal | medium yellow      | Efficient corrector — low FASR lifted by early-retry successes.                       |
| Bottom-left                   | small red          | No capability — FASR and IWSR near zero, SR low.                                      |
| Top-left, large dot, low FASR | large green/yellow | Dangerous case — high SR/IWSR mask retry dependence; SR misleads without FASR.        |



### A.3.4 Execution and Reasoning

Executability ratio, Precondition Awareness Score, the sequencing-vs-fabrication split, and mean temporal distance — everything the PDDL Simulator extracts by replaying a plan step by step.

These metrics look *inside* a plan, valid or not, by replaying it through the `PDDLSimulator`. They are the framework's answer to "a plan failed — but how close was it, and what kind of mistake was it". All are produced together by `compute_precondition_metrics`, which walks the plan step by step until the first precondition failure.

#### How one failure is classified

At the first failing action the simulator inspects each unsatisfied propositional precondition and asks

`last_step_atom_true(atom)`

against its per-step history

`prop_hist`

. If the atom was true at some earlier step →

**sequencing error**

(the model deleted a prerequisite too early). If it was never true →

**state fabrication**

(the model invented a state that never existed). A wrong-arity action is counted as fabrication and halts the walk.

#### METRIC · ER / Exec: Executability Ratio

$exec\_ratio = first\_failure\_index / plan\_length$  How far into the plan the model gets before the first illegal action — a continuous measure of partial correctness that is completely independent of whether the goal is reached.

**How it is computed** The simulator applies effects step by step; at the first unsatisfiable precondition (or wrong arity) it records `first_failure`. A fully executable plan yields ratio 1.0 even if it never reaches the goal.

```
for step, action in enumerate(actions):
    ok, failed_prop, failed_num = sim.check_preconditions(...)
    if not ok:
        first_failure = step; break
    sim.apply_effects(...)
return {"executability_ratio": first_failure / plan_len}
```

**Range & meaningful bands** NaN for empty plans / missing PDDL. Bands: < 0.30 No-Grounding · 0.30–0.70 Vocabulary-Only · > 0.70 Understander (runs deep, still fails the goal). In the bundled run kimi reaches 0.969 and gpt\_oss 0.952 — both produce structurally near-perfect plans.

**In the plots** `executability_by_model_domain` (boxplots), `executability_vs_length` (faceted scatter—testing whether executability decays with plan length), and the radar/ranking views, where it is treated as a max-direction metric.

**Rationale** Huang 2022 established executability and goal-correctness as independent axes. A high exec with low SR is the Understander/Vocabulary signature: the model speaks the language and respects local preconditions but cannot sequence to the goal.

#### METRIC · PAS: Precondition Awareness Score

$PAS = seq\_errors / (seq\_errors + fab\_errors)$  Of the failures the model makes, what fraction are *ordering* mistakes (it understood the state model but mis-ordered) versus *fabrication* (it invented impossible states)c

**How it is computed** Computed at the single failure point. Each failed propositional precondition is bucketed via the temporal lookup; PAS is the sequencing share of all failures.

```
for atom in failed_prop:
    if sim.last_step_atom_true(atom) >= 0: seq_errors += 1    # was true before
    else: fab_errors += 1    # never true
fab_errors += len(failed_num)
PAS = seq_errors / (seq_errors + fab_errors)    # NaN if no failures
```

**Range & meaningful bands** NaN when the plan fully executes (no failures) → substituted with a 0.5 prior so a perfect model is not penalised. High PAS + low SR = Understander; low PAS = Vocabulary-Only / incoherent world model.

**In the plots** The y-axis of `failure_mode_taxonomy`, a radar axis, and a ranked metric. The raw counts

also drive `failure_type_breakdown`.

**Rationale** This is the metric that distinguishes Dziri’s compositional collapse (sequencing) from Vafa’s world-model incoherence (fabrication). Two models with identical SR and exec can sit at opposite ends of PAS and require completely different interventions.

### A.3.5 Sequencing vs fabrication — the failure breakdown

`failure_breakdown` sums the two error counts per model and reports their proportions. In the bundled run almost every model is fabrication-dominant (proportion  $\approx 1.0$ ); the exception is gemma\_4\_31b at 0.42 sequencing / 0.58 fabrication — the only model in the set making a substantial share of "right idea, wrong order" mistakes rather than inventing states outright.

| Model        | seq | fab | seq prop. |
|--------------|-----|-----|-----------|
| gemma_4_31b  | 20  | 28  | 0.417     |
| qwen_3_6_27b | 5   | 19  | 0.208     |
| nemotron(hf) | 3   | 28  | 0.097     |
| phi_4        | 1   | 32  | 0.030     |
| gpt_oss_120b | 0   | 1   | 0.000     |

#### Why strong models show tiny failure counts

gpt\_oss and kimi show only 1 total failure each — not because they fail gracefully, but because they *succeed* on most instances, so there is rarely a first-failure to classify. The failure breakdown is informative only where failures are plentiful; on near-solving models its proportions are statistically meaningless. Read it together with SR, never alone.

#### METRIC · $\tau$ : Mean Temporal Distance

$\tau = \text{mean}(\text{failure\_step} - \text{last\_true\_step})$  For sequencing errors only: how many steps back the violated prerequisite was last true. A measure of *how badly* the ordering collapsed.

**How it is computed** For each sequencing error the gap between the failure step and the last step the atom held is recorded;  $\tau$  is the mean of those gaps.

```
last_step = sim.last_step_atom_true(atom)
temporal_distances.append(step - last_step)
mean_temporal_distance = np.mean(temporal_distances)
```

**Range & meaningful bands** Unbounded above; NaN when there are no sequencing errors (i.e. fabrication-only or fully-executing models — most of the bundled set). Small  $\tau \rightarrow$  the prerequisite was deleted just before it was needed (a local slip). Large  $\tau \rightarrow$  it was deleted many steps earlier (global ordering failure).

**In the plots** `temporal_distance_by_model` (per-model histograms). A ranked metric (direction min).

**Rationale** Refines PAS: among models that mostly make sequencing errors,  $\tau$  says whether they are nearly-right (short  $\tau$ ) or have lost the plot (long  $\tau$ ) — the granular read on long-horizon compositional ability.

### A.3.6 Reading the execution plots together

These four figures are meant to be read as a cascade, not in isolation. `executability_ratio` is a prerequisite for PAS to mean anything: a model with exec near 0 has nothing to classify — it fails before any causal structure can be tested, so PAS is almost always NaN there. PAS becomes informative only once exec is moderate-to-high ( $\approx 0.4+$ ), i.e. once the model has actually engaged the domain’s state-transition logic long enough for its causal reasoning to be observable.

**Executability ratio by model  $\times$  domain (box plot).** Box edges are Q1/Q3, the middle line the median, whiskers  $1.5 \times \text{IQR}$ , dots beyond them outliers. A box near 1.0 means the model runs most of the plan before any failure (it understands domain structure); a box at 0.0–0.2 means it fails at step 1–2 (near-total incomprehension). A wide IQR means it works on some problems and collapses on others; a narrow IQR means consistency.

The ideal is a **narrow box sitting high across every domain**— consistently good, most plans run before failing, little variation.

Cross-comparisons follow from the box meaning: comparing differently-coloured boxes *within* one domain cluster reads one model’s behaviour across domains (its reasoning across scenarios); comparing the same colour *across* domains reads each domain’s real difficulty.

**Failure-type breakdown per model (stacked bar).** Raw errors are summed across all domains/problems then converted to proportions (each row divided by its total), so models with different run counts stay comparable; a model with zero failures gets a flat zero bar rather than NaN.

| Segment                  | Meaning                                                                                                                                                                     | Severity                                |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Blue — sequencing</b> | The required atom was true earlier but the action was placed too late, after a prior effect deleted it. A tall blue bar = a reasonable causal model with ordering mistakes. | More fixable with better prompting      |
| <b>Red — fabrication</b> | The required atom was never true in any prior state — the model invented a precondition. A tall red bar = no grounded world model.                                          | Severe; less fixable by prompting alone |

#### Atom vs precondition

A precondition is a conjunction of several *atoms* (single grounded propositional facts). The sequencing/fabrication verdict is made per failed atom at the first failing action, which is why a single action can contribute both kinds of error.

**Executability ratio vs plan length (faceted scatter).** The decay slope is the diagnostic. A negative correlation between plan length and `exec_ratio` is exactly Dziri et al.’s prediction (NeurIPS 2023): transformers lose causal coherence as the computation graph deepens. Faceting by domain separates a genuine state-tracking limit from a domain-specific one.

| Pattern                              | Reading                                                                                                                                                                                              |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exec stays high regardless of length | Robust internal model.                                                                                                                                                                               |
| Exec drops as length grows           | State-tracking decay — the model leans on its own generated chain more than a strong internal representation. (A natural place for a Vafa-style Myhill-Nerode world-model metric, generalising PAS.) |
| Flat near 0 everywhere               | Little-to-no domain knowledge, or the domain was never read.                                                                                                                                         |
| Decays in <i>every</i> domain        | Fundamental state-tracking capacity problem.                                                                                                                                                         |
| Decays in <i>one</i> domain only     | Domain-specific problem.                                                                                                                                                                             |

**Mean temporal distance per model (histogram).** For a sequencing error, distance = step of first failure—last step where the violated atom was true, averaged across the atoms of the failing precondition. Each bar groups plans whose sequencing errors had that mean distance. A model whose `mean_temporal_distance` is always NaN produces *no panel*—all its failures are fabrications, there are no sequencing errors to measure, and the absence itself is informative: it means PAS = 0.0.

| Distribution shape | Interpretation                                                         | Severity |
|--------------------|------------------------------------------------------------------------|----------|
| Spike at 1–2       | Near-miss ordering — almost right, one step too late.                  | Low      |
| Spike at 3–7       | Imprecise causal model — knows what’s needed, tracks sequence loosely. | Moderate |
| Long tail 8+       | Severe ordering collapse — preconditions were true much earlier.       | High     |
| No bars (NaN)      | All failures are fabrications — no causal knowledge at all.            | Severe   |

#### The cascade — read all four with one question chain

(1) Does the plan run at all (box plot)c (2) When it fails, is it conceptual or ordering (breakdown)c (3) Does the failure worsen with complexity (exec-vs-length)c (4) How severe are the ordering errors (temporal distance)c A model that passes (1), is mostly blue in (2), stays flat in (3) and spikes at distance 1 in (4) genuinely understands the domain and makes only minor sequencing slips. A model that fails (1) and has no histogram in (4) is fabricating preconditions from the first step — it may not be planning at all.

### A.3.7 Grounding and Discriminatory

Action, fuzzy, and object hallucination, plus the inverse rate used downstream — the metrics that decide whether a model is reading the formal domain at all.

Hallucination is the most basic grounding signal: is the model even using the domain's legal vocabulary? A model that hallucinates is not reading the schema at all; a model that does not hallucinate has at minimum memorised it, even if it cannot sequence. Three variants separate three failure modes, all produced by `compute_hallucination_metrics` in one pass.

#### Why "discriminatory"

These metrics rarely correlate with success on their own — most capable models score 0 hallucination — but they are decisive at the *boundary*.

. They are what tells apart a Vocabulary-Only model (legal names, no coherent state) from a No-Grounding model (invents names outright). In the bundled run hallucination is 0 for almost every model; the exceptions are phi\_4 (0.120) and qwen\_3\_6\_27b (0.143).

#### METRIC · Halluc: Action Hallucination Rate

*rate = illegal\_action\_names / total\_actions* Fraction of plan steps whose action name is not in the domain's legal action set. The headline grounding gate.

**How it is computed** Each parsed action name is checked for membership in the action-name set stored in `domain_info` under the key `action_names`.

```
if action_name not in legal_actions:
    hall_strict += 1
hallucination_rate = hall_strict / total_actions # NaN if empty
```

**Range & meaningful bands** NaN for empty plans / absent domain file. The profile gate is strict: IHR > 0.90 (i.e. hallucination < 0.10) is required for the grounded profiles; IHR < 0.60 marks No Grounding.

**In the plots** `hallucination_heatmap` (model×domain), `hallucination_by_model_domain` (bars), the x-axis of `failure_mode_taxonomy`, and a ranked metric (direction min).

**Rationale** Valmeekam's obfuscation result (35%→0.16% when action names are scrambled) shows that apparent planning can collapse to nothing once the model can no longer pattern-match on familiar names. Hallucination rate is the everyday proxy for that grounding.

#### METRIC · Fuzzy: Fuzzy Hallucination Rate

*counted only if min Levenshtein > 2* Strict hallucination minus the near-misses. An illegal name is counted here only if its edit distance to *every* legal name exceeds 2, so "pick-up" vs "pickup" is forgiven but a genuine invention is not.

**How it is computed** Among strict hallucinations, only those far from any legal name survive.

```
if min(levenshtein(action_name, a) for a in legal_actions) > MAX_FUZZY_DISTANCE: # =2
    hall_fuzzy += 1
```

**Range & meaningful bands** Always ≤ strict rate. The *gap* between strict and fuzzy is the diagnostic: a model with high strict but low fuzzy is making transcription/format slips, not inventing actions — a parser/formatting problem, not a grounding problem.

**In the plots** Carried in the aggregate tables (`Fuzzy_Halluc`); read alongside the strict heatmap rather than plotted on its own.

**Rationale** Separates "the model doesn't know the action exists" from "the model knows it but spelled it differently". Only the former is a reasoning failure; the latter is fixable at the parser/prompt level and should not be charged against planning ability.

#### METRIC · ObjHalluc: Object Hallucination Rate

*rate = illegal\_arg\_tokens / total\_arg\_tokens* Fraction of argument tokens not present in the problem's `:objects` block — normalised over arguments, not actions. Catches models that know the verbs but invent the nouns.

**How it is computed** Every argument of every action is checked against `problem_info["objects"]`.

```

for arg in action_args:
    if legal_objects and arg not in legal_objects:
        obj_hall += 1
object_hallucination_rate = obj_hall / total_args

```

**Range & meaningful bands** An independent axis from action hallucination. High object-halluc + low action-halluc = the model has the action vocabulary but invents world entities — shallow grounding, exactly Vafa’s incoherent-world-model signature at the object level.

**In the plots** `object_hallucination_by_model_domain` (bars).

**Rationale** Object grounding and action grounding fail independently. Splitting them prevents a model that uses legal actions on fabricated objects from looking grounded.

#### METRIC · IHR: Inverse Hallucination Rate

$IHR = 1 - hallucination\_rate$  The same signal flipped so higher is better, for use in the composite score and the profile thresholds.

**How it is computed** `1 - hallucination_rate`, NaN-preserving.

```

inverse_hallucination_rate = 1 - hallucination_rate

```

**Range & meaningful bands** IHR is used as a radar axis and as a composite component (weight 0.20, via `one_minus_halluc`). Profile gates:  $IHR > 0.90$  indicates grounded behaviour, whereas  $IHR < 0.60$  indicates the *No Grounding* profile.

**In the plots** Radar axis (`radar_chart`); component of the Composite Planning Score.

**Rationale** Purely a sign convention so every composite input points the same direction (higher = better) and the weighted sum stays interpretable.

#### NaN handling differs by consumer

In the aggregate tables hallucination is averaged NaN-aware. But inside the per-row bootstrap for the composite CI, a NaN hallucination is replaced by a neutral 0.5 for

`one_minus_halluc` rather than dropped. This is deliberate (an empty plan shouldn’t score as perfectly grounded) but it means the composite CI and the table mean can diverge for models with many empty plans — see section 06.

### A.3.8 CoT Alignment (CoTaL)

From a capped text-overlap proxy to a five-component structural comparison of the emitted plan against the plan reconstructed from the reasoning trace — plus the status taxonomy that says how far to trust each score.

CoTaL — the CoT plan-alignment score — answers the question Lanham 2023 and Kambhampati 2024 pose: *does the model's reasoning actually correspond to what it does, or is the chain-of-thought a post-hoc story?* The metric began as a crude text-overlap proxy and was redesigned into a structural sequence comparison between two reconstructed plans. Both still exist in the code, but they play different roles: the structural score is the real metric; the text proxy is a capped fallback for when there is only one plan to look at.

#### The two-plan idea

For a reasoning model, the parser produces

**two**

plans per attempt:

`raw.actions`

(what the model emitted in its answer) and

`reasoning.actions`

(the plan reconstructed from the thinking trace). CoTaL is, at its strongest, the structural agreement between these two sequences. High agreement = the model did what it reasoned about (faithful CoT). Low agreement = the answer and the reasoning diverged.

### A.3.9 The status decision tree

`compute_cot_alignment_for_attempt` routes each attempt into one of several statuses depending on which plans exist and whether they validate. The status is as important as the number — it tells you how much to trust the score.

| Status                                            | Condition                          | Score basis                    | Confidence |
|---------------------------------------------------|------------------------------------|--------------------------------|------------|
| <code>comparable_and_both_valid</code>            | raw + reasoning plans, both valid  | structural_-alignment          | high       |
| <code>comparable_but_both_invalid</code>          | both plans present, both invalid   | structural_-alignment          | high       |
| <code>comparable_but_raw/reasoning_invalid</code> | both present, one invalid          | structural_-alignment          | high       |
| <code>semantic_proxy_only</code>                  | raw plan but no reasoning plan     | text overlap $\times$ 0.35 cap | low        |
| <code>no_raw_plan</code>                          | reasoning but no emitted plan      | none                           | low        |
| <code>no_reasoning_text</code>                    | neither reasoning text nor actions | none                           | none       |

### A.3.10 The strict metric — structural\_alignment

When both plans exist, `compute_sequence_alignment` (in `evaluators/plan_alignment.py`) compares the two normalised action sequences along five order- and content-sensitive dimensions and combines them into one bounded score:

$$\text{structural alignment} \in [0, 1] = 0.35 \cdot \text{lcs\_ratio} + 0.25 \cdot \text{prefix\_ratio} \\ + 0.20 \cdot \text{bag\_overlap} + 0.10 \cdot \text{length\_ratio} + 0.10 \cdot \text{repetition\_similarity}$$

Alongside the score it returns diagnostics for the report: `exact_sequence_match`, `first_mismatch_index`, `adjacent_swaps` (transposition localisation), `displaced_actions`, per-sequence distinct-action counts, action frequencies, and run-length `repetition_blocks`. These are exactly the distinct/repetition/transposition diagnostics the redesign called for.

### A.3.11 The fallback — semantic proxy, capped at 0.35

When only the raw plan exists, with no reconstructable reasoning plan, the metric falls back to "`compute_cot_semantic_support`": the fraction of the action names and object names in the plan that appear as tokens in

| Component             | What it captures                                                | Why weighted so                                                     |
|-----------------------|-----------------------------------------------------------------|---------------------------------------------------------------------|
| lcs_ratio             | longest common subsequence / max len — order-preserving overlap | highest (0.35): the truest measure of "same plan in the same order" |
| prefix_ratio          | common leading actions / min len — do they start identically?   | 0.25: a shared prefix is strong evidence of a shared plan           |
| bag_overlap           | multiset intersection — same actions, order ignored             | 0.20: same ingredients even if reordered                            |
| length_ratio          | min/max length — are the plans the same size?                   | 0.10: weak signal, easily coincidental                              |
| repetition_similarity | similar amount of repeated actions                              | 0.10: catches degenerate looping                                    |

the reasoning text. This is vocabulary coverage, not plan agreement, so it is deliberately **capped at 0.35**—it can never masquerade as a strong structural match.

```
score = semantic["cot_semantic_support_score"]          # (action_cov + object_cov)/2
proxy_score = score * semantic_proxy_cap                # cap = 0.35
status = "semantic_proxy_only"; confidence = "low"
```

#### Why the cap matters

Mentioning the right words is not the same as planning them in the right order. The cap encodes that epistemic gap: a proxy score of 0.35 is the ceiling of "the reasoning at least talks about the plan", which is strictly weaker than "the reasoning *is* the plan".

#### A.3.12 Which plan is the basis, and which attempt is selected

When both plans validate-differently, a three-way rule picks the authoritative one: prefer the valid plan if only one is valid; if both are valid prefer the *shorter*; if both invalid prefer the reasoning plan. Across iterations, `select_cot_alignment_attempt` takes the first solved attempt when the instance is solved, otherwise the last attempt — so the reported CoTaL reflects the decisive attempt, not an average of doomed ones.

```
if raw_valid and not reasoning_valid: basis = "raw"
elif reasoning_valid and not raw_valid: basis = "reasoning"
elif raw_valid and reasoning_valid:    basis = "raw" if len(raw) <= len(reasoning) else "reasoning"
else:                                  basis = "reasoning"
```

#### A.3.13 Aggregation & range

The `overall` table actually exposes three CoT aggregates: `strict_mean_cot_plan_alignment` (structural only), `inclusive_mean_cot_alignment` (strict-or-proxy), and `mean_cot_semantic_support` (raw coverage), plus status-rate columns like `cot_comparable_and_both_valid_rate` and `cot_exact_sequence_match_rate`. Report the strict mean as the metric and the status rates as the coverage context.

#### The single most important caveat for this metric

CoTaL is only defined where paired plans (or at least reasoning text) exist. In the bundled run the status distribution is

**282 no\_reasoning\_text · 82 semantic\_proxy\_only · 18 comparable\_and\_both\_valid**  
— most instances have

*no*

reasoning plan to compare, so `CoT_Alignment` is NaN for most non-reasoning models. Where it is defined on tiny samples it can be misleading: kimi shows `CoT_Alignment`

=



1.000 driven by a handful of comparable attempts while its SR is only 0.051. Never read a CoTaL mean without its sample size and status mix; a high CoTaL on a low-SR model is the textbook "the reasoning matches a plan that still fails" case, not evidence of capability.

#### Known accessor edge case

`parsed_plan_reasoning_text()`  
 returns the reasoning field only when it is a string; for the current dict-shaped  
`reasoning`  
 namespace it returns `""`. In practice the reasoning text is recovered first from the raw generation via  
`raw_attempt_reasoning_text()`  
 , so the proxy path still fires — but if a run lacks raw generation text, the proxy will silently see an empty string. Worth gating with an explicit diagnostic.

### A.3.14 The coverage proxy: the double-intersection

Underneath the structural metric sits the semantic-support proxy — the layer that answers the plain question *"does the chain-of-thought actually talk about the same things that end up in the plan?"* If the CoT names the right actions and objects, that is evidence the reasoning guided generation (high score); if it talks about entirely different things, the reasoning is decorative and was likely written around a plan produced by some other mechanism (low score). This operationalises Kambhampati's "fake reasoning vs genuine deliberation" and Lanham et al.'s finding that CoT grows less faithful with scale.

Coverage is computed with a deliberate **two-step filter**:

Both filters are necessary. **Without the first**, any English word resembling a PDDL action name would inflate the score; restricting to exact members of `legal_action_names` stops that. **Without the second**, a CoT that simply lists every action in the domain would score 1.0 even if the plan used one action; requiring the specifically-used actions stops that.

#### Why the denominator is the plan, not the CoT

Each coverage score divides by the size of the  
*plan*  
 vocabulary, not the CoT vocabulary. The question is "what fraction of what the model  
*did*  
 is reflected in its reasoning?", not "what fraction of the reasoning made it into the plan?". A plan-based denominator rewards coverage of what was actually done and denies a long, everything-mentioning CoT an inflated score.

**Worked example.** CoT: *"move car1 from l1 to l2 along the road; once car1 is at l2, pick up passenger p1; then drive to l3 to drop off the passenger."* Plan: (move-car car1 l1 l2) (pick-up p1 car1 l2) (move-car car1 l2 l3) (drop-off p1 car1 l3). Every plan action and object is named in the prose, so both action- and object-coverage are full.

| Scenario                                      | action_cov | object_cov | What it reveals                                                   |
|-----------------------------------------------|------------|------------|-------------------------------------------------------------------|
| CoT names every action and object in the plan | 1.00       | 1.00       | Perfect alignment — the CoT anticipated the plan.                 |
| All actions, wrong objects                    | 1.00       | 0.00       | Knows what to do, reasons about the wrong objects.                |
| Some actions, all objects                     | 0.50       | 1.00       | Object tracking good, action selection partly off.                |
| No domain actions or objects                  | 0.00       | 0.00       | CoT is pure prose with no PDDL grounding.                         |
| Empty plan / empty CoT                        | 0.00       | 0.00       | $\max(0,1)$ denominator avoids division by zero $\rightarrow 0$ . |

#### Tokeniser is substring-based, not word-boundary

Tokenisation uses

```
re.findall(r'[a-z][a-z0-9_-]*', cot_lower)
```

, which matches substrings rather than whole words. If the CoT contains "pick-up-person" and the action is "pick-up", the shorter token still matches, so the score can be slightly inflated when an action name is a substring of a longer compound word. Rare for specific PDDL action names, but worth remembering when reading very high scores.

#### A.3.15 Reading the CoT plots

**Mean CoT alignment by model  $\times$  domain (bars).** The mean hides variance, so it is the first thing to look at before the violin. A mean of 0.60 could be every plan near 0.60, or half at 1.0 and half at 0.2 — the violin resolves it; read both. Across a model's domains: similar-and-high means domain-general grounding (formal vocabulary used regardless of domain); dissimilar means domain-specific behaviour (grounded in some domains, lapsing into prose in others — a transfer failure); similar-and-low means the CoT is not grounded in PDDL vocabulary at all. Comparing one domain across models yields a qualitative sense of how hard that domain's names/identifiers are for CoT to latch onto.

| Pattern                  | Conclusion                                                                                                                                                                                                          |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CoT=True > False         | CoT is causally helping — <i>but</i> only counts as genuine reasoning if the alignment score is also high; if success rises while alignment is low, the CoT is not what improved the outcome (Stechly et al. 2023). |
| CoT=True < False         | CoT is actively harmful — verbose self-contradicting reasoning confuses plan generation.                                                                                                                            |
| CoT=True $\approx$ False | CoT is decorative — Kambhampati's "fake reasoning": plans retrieved by pattern-matching, CoT written around them.                                                                                                   |
| Both low                 | Model cannot plan here regardless; the CoT effect size is not interpretable at the floor.                                                                                                                           |

#### Success rate: CoT=True vs CoT=False (paired bars).

**This contrast is only valid if everything else is held equal**

The CoT=True/False runs must test the same problems under the same prompt, **temperature** and **iteration budget**, or the bar difference reflects confounds rather than a CoT effect. Temperature divides the logits before sampling (adjusted = score/temperature): < 1 sharpens toward the most likely tokens, > 1 flattens toward surprising ones — so a reproducible benchmark should pin temperature to 0 or near-0 for every model so differences reflect capability, not sampling luck. The iteration budget (N rounds) follows Valmeekam et al. 2023 for the cap and Stechly et al. 2023 for treating attempts past the first as independent samples; 3–15 is the defensible range since the marginal value of later iterations is negligible. Both must be identical across the True and False runs.

**Alignment distribution for valid vs invalid plans (violin).** Compare the median lines between the Valid and Invalid violins: a substantially higher Valid median means alignment is predictive of success (models reasoning in domain vocabulary are likelier to succeed); equal medians mean alignment and validity are independent. The violin is a 90°-rotated, KDE-smoothed histogram (x = alignment score, y = plan count) — and because CoTaL scores cluster at the extremes, the KDE is imperfect and should be read loosely.

| Valid violin              | Invalid violin                | Conclusion                                                                                                                     |
|---------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Tight, high mass near 1.0 | Wide, spread 0–1              | Valid plans have consistently aligned CoT; failed plans have chaotic CoT — no reasoning $\leftrightarrow$ output relationship. |
| Moderately tight, high    | Slightly lower, similar width | CoT stays grounded even in failures — failures are sequencing/complexity, not reasoning misalignment.                          |
| Very wide, mid-range      | Wide, similar                 | CoT noisy for both outcomes — alignment is essentially random.                                                                 |

### A.3.16 PS (Composite Planning Score)

What PS represents, its five weighted components plus the conditional CoT bonus, the per-row bootstrap CI, how to read the capability radar, and how to retune the weights for a specific research goal.

The Composite Planning Score collapses five orthogonal capability axes into one number per model (and per model×domain). Its purpose is triage, not verdict: a single ranking to scan, always reported alongside its components because a composite can hide compensating strengths and weaknesses.

### A.3.17 Why a composite at all

The preceding sections give nine-plus dials. PS exists so that, for a quick comparison, you have one axis that already encodes the framework's value judgement: zero-shot ability matters most, retries are a cost, structural quality and grounding matter, and precondition awareness is a tie-breaker. It is the metric you sort the leaderboard by; it is never the metric you conclude from.

### A.3.18 The components and their weights

PS is a weighted sum of five inputs, each already in [0,1], with an optional sixth (CoT) bonus. The person sometimes thinks of it as "two components" — that intuition maps onto the structure exactly: there is a **base** (the five-term weighted sum) and a **conditional CoT bonus** that only applies when a strict CoT alignment score exists, at which point all weights are renormalised so the result stays bounded.

$$\begin{aligned} \text{base} &= 0.25 \cdot \text{FASR} + 0.20 \cdot \text{IWSR} + 0.20 \cdot \text{exec\_ratio} + 0.20 \cdot (1 - \text{halluc}) + 0.15 \cdot \text{PAS} \\ \text{PS} &= \frac{\text{base} + 0.05 \cdot \text{CoT}}{1.05}, \quad \text{if CoT is finite} \\ &\rightarrow \text{renormalised and clipped to } [0, 1] \end{aligned}$$

| Component  | w     | Rationale for the weight                                                | NaN substitute |
|------------|-------|-------------------------------------------------------------------------|----------------|
| FASR       | 0.25  | Highest — the only signal that cannot be inflated by retries            | 0              |
| IWSR       | 0.20  | Retry efficiency; rewards fast convergence, penalises stochastic search | 0              |
| exec_ratio | 0.20  | Structural quality independent of goal success                          | 0              |
| 1 - halluc | 0.20  | Schema grounding                                                        | 0.5            |
| PAS        | 0.15  | Lowest — undefined for zero-failure models, so a 0.5 prior is needed    | 0.5            |
| CoT bonus  | +0.05 | Applied only when strict CoT alignment is finite                        | skipped        |

### A.3.19 Aggregation & range

PS is computed on the *aggregated* components in both the `overall` and `by_domain` tables — so each model carries an overall PS and one PS per domain. In the bundled run: gpt\_oss 0.548 · nemotron(nv) 0.446 · kimi 0.435 · qwen\_3\_5 0.425, down to phi\_4\_mini 0.315.

### A.3.20 The 95% bootstrap confidence interval

To show whether a PS difference is real or noise, the `composite_score` table attaches a bootstrap CI (N=1000, seed=42). Crucially it resamples **per-row PS contributions**, not per-model means — a more conservative estimate that reflects instance-level variability.

```
contribs = model_df.apply(per_row_composite, axis=1).values
boots = [contribs[rng.integers(0, len(contribs), len(contribs))].mean() for _ in range(1000)]
PS_ci_lo, PS_ci_hi = np.percentile(boots, 2.5), np.percentile(boots, 97.5)
```

#### Overall PS and the CI are computed two different ways

The overall PS averages the components first, then weights them. The CI averages per-row composites. These are not the same operation — for non-linear cases and for models with many empty/NaN rows they can diverge, and the overall PS can even sit outside its own CI (in the bundled run kimi's PS\_overall

0.435 lies above its CI [0.184, 0.247]). Read the CI as a stability indicator for the per-instance score, not as a strict interval around PS\_overall. If you need them to coincide, compute the point estimate the same per-row way.

### A.3.21 The capability radar

The `radar_chart` plots the five [0,1] axes **FASR** · **IWSR** · **Exec** · **IHR** · **PAS** as a pentagon per model. It is the visual companion to the profile thresholds: a Genuine Planner fills most of the pentagon; a No-Grounding model collapses to the centre; a Vocabulary-Only model shows a lopsided shape — high IHR but low PAS and Exec. Read *area* for overall strength and *shape* for the capability signature. Note the radar uses IHR (not 1-halluc) and omits the CoT bonus, so it is the *base* profile, not the bonus-adjusted PS.

#### Radar reading caveat

Equal area can come from very different shapes — a model strong on FASR/IWSR but weak on Exec/PAS can match the area of a balanced model. Always compare shape, not just fill. With many models the overlapping fills become unreadable; facet or pick a handful for any figure you publish.

### A.3.22 Goal-dependent weight tuning

The weights encode a scientific claim; changing them changes the question PS answers. They live in one dict (`COMPOSITE_WEIGHTS`) and must sum to 1.0. Reasonable retunings for different research goals:

| Research goal             | Adjustment                       |
|---------------------------|----------------------------------|
| Zero-shot capability only | FASR→0.40, IWSR→0.10             |
| Domain grounding focus    | one_minus_halluc→0.35, FASR→0.15 |
| All dimensions equal      | all five = 0.20 (simple average) |
| Execution-centric         | exec_ratio→0.35, PAS→0.10        |

#### The discipline that makes PS safe

Always report the component scores next to PS. A PS of 0.60 from FASR=1.0 / exec=0.0 (always right on the first try, never produces a partially-executable failure) is a completely different model from a PS of 0.60 from moderate scores everywhere. The composite ranks; the components explain.